

Geometrically Nonlinear Effects in Wing Aeroelastic Dynamics at Large Deflections

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Abstract

This paper investigates geometrically nonlinear effects due to nonlinear kinematics and follower aerodynamics in the aeroelastic dynamics of a very flexible wing. The test case is the Pazy wing, a benchmark model for geometrically nonlinear aeroelastic studies involving large deflections in low-speed flow. The work builds on a low-order model of the wing composed of a geometrically nonlinear beam coupled with potential flow thin airfoil theory. Nonfollower aerodynamics underestimates static deflections by up to 10% while linear kinematics overestimates them by up to 50%. At low static deflections, the wing hump mode flutter mechanism is driven by three-dimensional unsteady aerodynamic effects. As static deflections increase, the flutter onset and offset speeds are driven by curvature effects that can only be captured by nonlinear kinematics, which reduce the natural frequencies associated with torsion and in-plane bending motions. Follower aerodynamics shrinks the hump mode instability region and shifts it to 1–4% lower flow speeds with no change in the flutter frequency, the qualitative trends, and the amplitude of limit-cycle oscillations. Overall, nonlinear kinematics is the primary geometrical nonlinearity influencing the wing aeroelastic behaviors: the modal and stability scenarios with and without follower aerodynamics collapse onto the same curves when presented as functions of the tip deflection. However, geometrical nonlinearities alone miss the subcritical nature of the hump mode flutter mechanism, which is attributed to aerodynamic stall and its interaction with large deflections. This work expands the growing literature on the aeroelastic dynamics of very flexible wings and will ease future studies considering nonlinear aerodynamic effects.

1. Introduction

The push for reducing aviation environmental impacts drives new aircraft designs toward lighter-weight wings with higher aspect ratios [1]. These very flexible wings experience large deflections under normal loads, questioning the validity of linear aeroelastic models and analysis methods commonly used in production environments [2]. This problem motivates research efforts to increase our fundamental understanding and predictive capabilities for very flexible wings to support the design of future energy-efficient aircraft.

The study of very flexible wings has spanned more than two decades. A significant effort has been devoted to exploring flutter instabilities and self-sustained oscillations (also known as limit-cycle oscillation), which are major aircraft design concerns [3]. Early numerical studies in very flexible, high-aspect-ratio wings highlighted that large static deflections alter the structural natural frequencies, consequently causing different flutter onset points and mechanisms than for the wing in its undeformed shape [4]. Experimental efforts further explored this problem, showing that very flexible wings may develop subcritical self-sustained oscillations [5] due to interactions between large structural deflections and aerodynamic stall effects [6]. Beyond root-clamped setups, researchers have also analyzed very flexible wings integrated into high-altitude long-endurance [7, 8, 9, 2] or blended-wing body aircraft [10]. These studies have revealed instabilities such as body-freedom flutter, which is driven by interactions between rigid-body and elastic motions in free flight, whose onset may also change with the wing static deflections. More recently, the push for sustainable commercial aviation has motivated similar studies in transport aircraft with very flexible, high-aspect-ratio wings [11], revealing trends in static response, natural frequencies, and flutter onset (among other metrics) similar to those previously observed in configurations for high-altitude long-endurance applications. The Third Aeroelastic Prediction Workshop (AePW3) Large Deflection Working Group (LDWG) is another recent effort aimed at advancing our understanding and prediction of very flexible wings. The group revolves around an experimental benchmark model for geometrically nonlinear aeroelastic studies in low-speed flow, the Pazy wing [12]. The unique feature of this wing compared with previous aeroelastic experiments [6] is its ability to withstand structural and low-speed aerodynamic loads causing tip vertical displacements up to 50% of the semispan. This feature has enabled static and flutter tests at unprecedented deflections [12, 13], providing new insights into large-amplitude aeroelastic

dynamics along with a rich experimental database for validating computational models across a spectrum of complexity [14, 15, 16, 17, 18, 19, 20, 21]. Due to its rising popularity, the Pazy wing has also been used as a test case for research on gust response [22], model-order reduction and active control for nonlinear aeroelastic systems [23, 24], and uncertainty quantification [25, 26], and other related topics.

Despite the growing literature on the aeroelasticity of very flexible wings, there remain many fundamental knowledge gaps in this field. One of these gaps revolves around the question: what is the impact of different nonlinear effects on wing aeroelastic dynamics at large deflections?

Earlier studies have attempted to answer this question. Tang and Dowell [6] investigated the interaction of geometrical structural nonlinearities and aerodynamic stall nonlinearities to understand the source of subcritical limit-cycle oscillations in their high-aspect-ratio wind-tunnel wing model. They attributed subcritical behaviors to stall, as these behaviors could not be predicted using numerical models without this effect. Stanford and Beran [27] conducted a similar assessment for the same wing in its undeformed shape, finding that structural nonlinearities played a destabilizing role. Patil and Hodges [28] investigated the role of structural and aerodynamic geometrical nonlinearities on the static response, natural frequencies, and flutter onset of a high-aspect-ratio wing. They found that structural nonlinearities due to large static deflections along with the geometrically exact calculation of the local angle of attack significantly affected the wing aeroelastic behavior, while aerodynamic nonlinearities due to non-planar wing and wake geometry played a secondary role. These studies provided valuable insights, but have some limitations. First, they did not consider wing static deflections beyond 20-25% semispan. Second, they considered only a handful of dynamic cases, such as a single flutter point, and did not extensively analyze a range of the flutter boundary, the complete stability scenario, or the post-flutter dynamics (limit-cycle oscillations). Finally, they did not analyze the impact of model complexity on the results. Howcroft et al. [29] explored this aspect by analyzing the impact of geometrical nonlinearities for a very flexible wing modeled using different approaches. However, the study only included static responses. Other researchers have investigated the aeroelastic behavior of geometrically nonlinear wings (e.g., see Refs. [4, 7, 8, 10, 9, 2, 11, 14]), but they have not isolated the impact of different nonlinear effects.

With the overarching goal of advancing the understanding of very flexible wings, this paper investigates geometrically nonlinear effects in wing aeroe-

lastic dynamics at large deflections. The specific objectives of the paper are: *i)* analyze the role of nonlinear kinematics and follower aerodynamics in the wing static aeroelastic response, natural frequencies in statically deformed shape, flutter boundary, and limit-cycle oscillations; and *ii)* to the extent where solutions from different models are available, assess how structural and aerodynamic model complexity impacts the predicted behaviors. The test case is the Pazy wing studied by the AePW3 LDWG. This wing is chosen for this study due to its publicly available properties, extensive experimental database, and ability to reach tip static deflections up to 50% semispan under representative structural and low-speed aerodynamic loads. Additionally, the wing exhibits multiple flutter mechanisms, including a hump mode associated with hysteresis—a phenomenon that has barely been explored in the literature on very flexible wings. Thus, this work makes new contributions to the field by exploring geometrically nonlinear effects over a wider deflection range than tackled in previous work [28] and by considering the impact of these effects on the flutter boundary, the complete stability scenario, and the limit-cycle oscillations associated with a subcritical hump mode flutter mechanism. To the author’s best knowledge, these new contributions are the first of their kind in the field.

The study builds on an existing low-order model of the Pazy wing, which was established by the authors in previous work [19, 18]. The model consists of a geometrically nonlinear beam described by a strain-based formulation [30] coupled with unsteady potential flow thin airfoil theory [31] corrected for wingtip effects. The model is implemented into the University of Michigan Nonlinear Aeroelastic Simulation Toolbox (UM/NAST) [10], a multi-disciplinary framework for modeling, analyzing, and simulating very flexible wings and aircraft. Static, modal, flutter, and limit-cycle oscillation analyses are conducted by considering multiple combinations of nonlinear or linear kinematics, with and without follower or nonfollower aerodynamics, to highlight the impact of different sources for geometrically nonlinear behavior. Results from the low-order model are compared with higher-order numerical solutions where possible.

The remainder of this paper is organized as follows: Sec. 2 describes the computational framework; Sec. 3 presents the wing benchmark case and its numerical model used in this work; Sec. 4 discusses the results; and a section of concluding remarks ends the paper.

2. Computational Framework

This work leverages UM/NAST [10], a low-order multidisciplinary framework for investigating very flexible wings and aircraft. While UM/NAST did not require any new development for this work, the formulation implemented into the framework is described below to make the paper self-contained and explain the modeling of geometrically nonlinear effects.

2.1. Theoretical Formulation

UM/NAST models an aircraft as a set of beams representing different components that undergo arbitrary deflections relative to a body-fixed frame $\mathbf{B}$, in rigid-body motion within an inertial frame $\mathbf{G}$. The $\mathbf{B}$ frame position in the $\mathbf{G}$ frame is given by the vector $\mathbf{p}_B$, which is resolved in the $\mathbf{B}$ frame, and its orientation is such that the unit vector $\mathbf{B}_x$ points downstream, $\mathbf{B}_y$ toward the right wingtip, and $\mathbf{B}_z$ completes the right-hand triad. The flexible motion is described by local frames $\mathbf{w}$, which origin at points $\mathbf{p}$ along each beam member and are such that $\mathbf{w}_x$ is along the beam reference axis, $\mathbf{w}_y$ is toward the leading edge, and $\mathbf{w}_z$ completes the right-hand triad. The deformed configuration of a generic point along the beam reference axis at a fixed time is described by the 12×1 vector

$$\mathbf{h}(s)^T = \{\mathbf{p}(s)^T \ \mathbf{w}_x(s)^T \ \mathbf{w}_y(s)^T \ \mathbf{w}_z(s)^T\} \quad (1)$$

In this equation, s is the curvilinear coordinate along the beam reference axis and $\mathbf{p}$, $\mathbf{w}_x$, $\mathbf{w}_y$, and $\mathbf{w}_z$ are 3×1 unit vectors resolved in the $\mathbf{B}$ frame.

The structural dynamics formulation implemented into UM/NAST is the geometrically exact, strain-based beam formulation of Su and Cesnik [30]. The main difference of this formulation compared with more widespread displacement-based formulations is that it assumes beam element strain measures as the independent structural degrees of freedom (DOF) instead of nodal displacements and rotations. Specifically, this strain-based formulation discretizes each beam member using three-node finite elements associated with four strain DOF: ε_x (axial extension), κ_x (torsion curvature), κ_y (out-of-plane bending curvature), and κ_z (in-plane bending curvature). The shear DOF are eliminated from the problem by condensing their effects through energy minimization [32], leading to a 4×4 cross-sectional constitutive law that relates the axial extension and the curvature strains for each beam element to the corresponding stress resultants over the cross section (axial force, twisting moment, and out-of-plane/in-plane bending moments).

Here, the term “out-of-plane bending” denotes bending in the plane defined by the local unit vectors $\mathbf{w}_x$ and $\mathbf{w}_z$ at each point along the beam reference axis while “in-plane bending” denotes bending in the plane defined by the local unit vectors $\mathbf{w}_x$ and $\mathbf{w}_y$. For a straight, flat, and initially untwisted beam in its undeformed shape, the planes where out-of-plane and in-plane bending occur coincide with the y - z and x - z planes of the body-fixed frame. This is no longer the case once the wing is deformed as the orientation of the local frame $\mathbf{w}$ changes along the beam reference axis according to its local curvature.

In the strain-based formulation of Ref. [30], the strain measures for a given beam element are assumed to be constant along its length, that is, each beam element introduces four strain DOF. Once the model strain vector is known, the deformed configuration is recovered as a dependent quantity by space marching appropriate kinematic relations given in Refs. [10, 30]. This process gives the h vector in Eq. (1) at each of the three nodes of each beam element (including repeated nodes at the coincident ends of consecutive beam elements). For aeroelastic calculations, the deformed configuration must be recovered at each numerical iteration (in static analyses) or time step (in dynamic analyses) to update the magnitude and orientation of the aerodynamic loads. In pure structural calculations where the applied loads are independent of the deformed configuration, this can be recovered at the end of the analysis from the converged strain-based solution.

The UM/NAST equations of motion are [10]

$$\begin{bmatrix} \mathbf{M}_{FF} & \mathbf{M}_{FB} \\ \mathbf{M}_{BF} & \mathbf{M}_{BB} \end{bmatrix} \begin{Bmatrix} \ddot{\boldsymbol{\varepsilon}} \\ \dot{\boldsymbol{\beta}} \end{Bmatrix} + \begin{bmatrix} \mathbf{C}_{FF} & \mathbf{C}_{FB} \\ \mathbf{C}_{BF} & \mathbf{C}_{BB} \end{bmatrix} \begin{Bmatrix} \dot{\boldsymbol{\varepsilon}} \\ \boldsymbol{\beta} \end{Bmatrix} + \begin{bmatrix} \mathbf{K} & 0 \\ 0 & 0 \end{bmatrix} \begin{Bmatrix} \boldsymbol{\varepsilon} \\ \mathbf{b} \end{Bmatrix} = \begin{Bmatrix} \mathbf{R}_F \\ \mathbf{R}_B \end{Bmatrix} \quad (2)$$

$$\dot{\boldsymbol{\zeta}} = -\frac{1}{2} \boldsymbol{\Omega}_{\boldsymbol{\zeta}} \boldsymbol{\zeta} \quad \dot{\mathbf{P}}_B = [\mathbf{C}^{GB} \quad 0] \boldsymbol{\beta} \quad (3)$$

$$\dot{\boldsymbol{\lambda}} = \mathbf{F}_1 \begin{Bmatrix} \ddot{\boldsymbol{\varepsilon}} \\ \dot{\boldsymbol{\beta}} \end{Bmatrix} + \mathbf{F}_2 \begin{Bmatrix} \dot{\boldsymbol{\varepsilon}} \\ \boldsymbol{\beta} \end{Bmatrix} + \mathbf{F}_3 \boldsymbol{\lambda} \quad (4)$$

The quantity $\boldsymbol{\varepsilon}$ is the model strain vector, $\boldsymbol{\beta}$ is the 6×1 vector of the rigid-body linear and angular velocities of the $\mathbf{B}$ frame with respect to the $\mathbf{G}$ frame (resolved in the $\mathbf{B}$ frame), and $\mathbf{b}$ is the 6×1 vector of their time integrals. The quantities $\mathbf{M}_{FF} = \mathbf{M}_{FF}(\boldsymbol{\varepsilon})$, $\mathbf{M}_{FB} = \mathbf{M}_{FB}(\boldsymbol{\varepsilon})$, $\mathbf{M}_{BF} = \mathbf{M}_{BF}(\boldsymbol{\varepsilon})$, and $\mathbf{M}_{BB} = \mathbf{M}_{BB}(\boldsymbol{\varepsilon})$ in Eq. (2) are the flexible, rigid-body, and coupling blocks of the generalized mass matrix of the model, that is, the mass matrix consistent with the flexible (strain) DOF and the rigid-body DOF (if

present); $\mathbf{C}_{FF} = \mathbf{C}_{FF}(\dot{\boldsymbol{\varepsilon}}, \boldsymbol{\varepsilon}, \boldsymbol{\beta})$, $\mathbf{C}_{FB} = \mathbf{C}_{FB}(\dot{\boldsymbol{\varepsilon}}, \boldsymbol{\varepsilon}, \boldsymbol{\beta})$, $\mathbf{C}_{BF} = \mathbf{C}_{BF}(\dot{\boldsymbol{\varepsilon}}, \boldsymbol{\varepsilon}, \boldsymbol{\beta})$, and $\mathbf{C}_{BB} = \mathbf{C}_{BB}(\dot{\boldsymbol{\varepsilon}}, \boldsymbol{\varepsilon}, \boldsymbol{\beta})$ are the corresponding blocks of the generalized damping matrix; and $\mathbf{K} = \mathbf{K}_{FF}$ is the stiffness matrix associated with the strain DOF of the strain-based formulation of Ref. [30]. The stiffness matrix is constant with deformation due to the small-strain assumption, which keeps the problem in the materially linear range (*i.e.*, linear stress-strain constitutive relation). Therefore, the present strain-based formulation contains a generalized stiffness matrix that is based on the linear constitutive relation between the beam strain measures and their corresponding stress resultants. The cross-sectional stiffness matrix for each beam element is only identified in a reference shape, such as the undeformed shape, and it is not updated during the solution process. Geometrically nonlinear effects associated with large deflections, which lead to nonlinear strain-displacement relations, are captured by updating the mass and damping matrices along with the vectors $\mathbf{R}_F = \mathbf{R}_F(\ddot{\boldsymbol{\varepsilon}}, \dot{\boldsymbol{\varepsilon}}, \boldsymbol{\varepsilon}, \dot{\boldsymbol{\beta}}, \boldsymbol{\beta}, \boldsymbol{\zeta}, \boldsymbol{\lambda})$ and $\mathbf{R}_B = \mathbf{R}_B(\ddot{\boldsymbol{\varepsilon}}, \dot{\boldsymbol{\varepsilon}}, \boldsymbol{\varepsilon}, \dot{\boldsymbol{\beta}}, \boldsymbol{\beta}, \boldsymbol{\zeta}, \boldsymbol{\lambda})$ in Eq. (2), which list the generalized loads associated with the flexible and rigid-body DOF. The quantity $\boldsymbol{\Omega}_{\zeta}(\boldsymbol{\beta})$ in Eq. (3) is the quaternion propagation matrix that relates the 4×1 quaternion vector $\boldsymbol{\zeta}$ to its time derivative $\dot{\boldsymbol{\zeta}}$, $\mathbf{C}^{GB}(\boldsymbol{\zeta})$ is the rotation matrix from the $\mathbf{B}$ to the $\mathbf{G}$ frame, and $\mathbf{P}_B = \mathbf{C}^{GB} \mathbf{p}_B$ is the position of the $\mathbf{B}$ frame within the $\mathbf{G}$ frame resolved in that frame. The vector $\boldsymbol{\lambda}$ lists the aerodynamic states of the theory of Peters et al. [31]. These states have the physical meaning of coefficients of a series expansion of the inflow due to the unsteady wake on the wing and are only defined for the beam elements associated with lifting members. These aerodynamic states capture unsteady effects due to wake vorticity shedding and convection downstream, and evolve in time according to the first-order equations (4) with coefficients given in the inflow matrices $\mathbf{F}_1 = \mathbf{F}_1(\boldsymbol{\varepsilon})$, $\mathbf{F}_2 = \mathbf{F}_2(\boldsymbol{\varepsilon})$, and $\mathbf{F}_3$. The matrices $\mathbf{F}_1$ and $\mathbf{F}_2$ are functions of the deformed shape and must also be updated during the solution to capture geometrically nonlinear effects [10].

UM/NAST implements multiple aerodynamic theories such as the potential flow thin airfoil theory of Peters et al. [31] used here. These theories compute the aerodynamic loads at the cross sections of the beam nodes based on local parameters such as the effective angle of attack, Mach number, and flap deflections. The local parameters are evaluated at the previous numerical iteration or time step and are mapped onto the aerodynamic loads using user-specified aerodynamic coefficient derivatives, lookup tables, or surrogate models. The aerodynamic loads may be corrected for wingtip, sweep, or compressibility effects. No displacement interpolation or load transfer scheme is

required because structural and aerodynamic information is exchanged at the beam nodes.

This work uses the UM/NAST static, modal, flutter, and dynamic solvers [10]. The static solver computes the deformed shape under aerodynamic and other applied loads. The modal solver computes the natural frequencies and mode shapes in the undeformed configuration or after a static solution by solving the generalized eigenvalue problem associated with the current generalized mass matrix and the constant stiffness matrix of the strain-based formulation. The flutter solver computes the stability scenario for a range of operating conditions by *i*) solving for the static deformed shape at each operating condition, *ii*) linearizing the equations of motion about each deformed shape, *iii*) extracting the eigenvalues of the local state matrix of each linearized system, *iv*) and processing the eigenvalues for all the operating conditions to identify instabilities and output root loci or frequency and damping diagrams. The dynamic solver first computes the static deformed shape at a given operating condition and then time marches the equations of motion to simulate the transient response for user-specified control inputs, gust disturbances, or other time-varying loads.

2.2. Geometrically Nonlinear Effects

This work investigates geometrically nonlinear effects associated with nonlinear kinematics and follower aerodynamic loads. UM/NAST can consider these effects individually or in combination, resulting in four cases: fully nonlinear, nonfollower loads, linear kinematics, and fully linear.

As mentioned in Sec. 2.1, UM/NAST uses beam strain measures as the independent structural DOF and recovers the deformed shape as a dependent quantity by space marching the strain-based kinematic equations [10, 30]. These equations are nonlinear by nature to capture shortening and curvature effects. In the linear kinematics case, the equations are linearized about the undeformed shape such that shortening and curvature effects are neglected. This results in a solution process akin to that of standard linear finite element solvers (except for using strain DOF in place of displacement DOF). Different from standard linear aeroelastic solvers, UM/NAST applies the aerodynamic loads in the deformed shape even in the case of linear kinematics.

To explain how UM/NAST accounts for follower aerodynamics, consider the potential flow thin airfoil theory of Peters et al. [31] with user-specified cross-sectional aerodynamic coefficient derivatives. With no control-surface

deflections, the local vertical force (lift) l , quarter-chord pitching aerodynamic moment $m_{c/4}$, and horizontal force (drag) d per unit length at a beam node are given by [33]

$$\begin{aligned} l &= \rho b \dot{y}^2 C_{l_\alpha} \alpha_{\text{eff}} + \pi \rho b^2 (-\ddot{z} + \dot{y} \dot{\alpha} - d \ddot{\alpha}) \\ m_{c/4} &= 2 \rho b^2 \dot{y}^2 (C_{m_0} + C_{m_\alpha} \alpha_{\text{eff}}) + \pi \rho b^3 \left[\frac{\ddot{z}}{2} - \dot{y} \dot{\alpha} - \left(\frac{b}{8} - \frac{d}{2} \right) \ddot{\alpha} \right] \\ d &= -\rho b \dot{y}^2 (C_{d_0} + C_{d_\alpha} \alpha_{\text{eff}}) \end{aligned} \quad (5)$$

Equation (5) involves the air density ρ , the local aerodynamic chord $c = 2b$, the airfoil properties C_{l_α} , C_{m_α} , C_{d_α} , C_{m_0} , and C_{d_0} , the beam node offset d from the quarter chord, the linear velocity components $\dot{y}$ (horizontal, upstream positive) and $\dot{z}$ (vertical, upward positive), the pitch rate $\dot{\alpha}$ (clockwise positive), the linear and angular accelerations $\ddot{z}$ and $\ddot{\alpha}$, and the effective angle of attack [33]

$$\alpha_{\text{eff}} = -\frac{\dot{z}}{\dot{y}} + \left(\frac{b}{2} - d \right) \frac{\dot{\alpha}}{\dot{y}} - \frac{\lambda_0}{\dot{y}} \quad (6)$$

where λ_0 is the zeroth-order term in the inflow expansion [31]

$$\lambda_0 \approx \frac{1}{2} \sum_{i=1}^{N_\lambda} b_i \lambda_i \quad (7)$$

In the original theory of Peters et al. [31], which considers a flat-plate airfoil immersed in the incompressible flow of an inviscid fluid, Eq. (5) becomes

$$\begin{aligned} l &= 2\pi \rho b \dot{y}^2 \alpha_{\text{eff}} + \pi \rho b^2 (-\ddot{z} + \dot{y} \dot{\alpha} - d \ddot{\alpha}) \\ m_{c/4} &= \pi \rho b^3 \left[\frac{\ddot{z}}{2} - \dot{y} \dot{\alpha} - \left(\frac{b}{8} - \frac{d}{2} \right) \ddot{\alpha} \right] \\ d &= -2\pi \rho b (\dot{z} + d \dot{\alpha} + \lambda_0)^2 \end{aligned} \quad (8)$$

The velocity and acceleration components in Eqs. (5) to (8) are in the local zero-lift aerodynamic frame at the current beam node [33]. In the follower case, the aerodynamic loads in Eq. (5) are defined in the same frame. For instance, the local force per unit length l is directed along the z axis of that frame, which coincides with the axis orthogonal to the chord line for a flat plate or another symmetric airfoil. The zero-lift aerodynamic frame at a

given beam node follows the structure in its deflection, and the resulting change in its orientation captures follower effects. In the nonfollower case, the aerodynamic loads in Eq. (5) are defined in the body frame such that, for instance, the local force l per unit length is always along the unit vector B_z . In either case, the aerodynamic loads are computed and applied in the current deformed shape considering the geometrically nonlinear local effective angle of attack in Eq. (6). Neglecting follower effects thus impacts the solution in the following ways: it misses the changes in the aerodynamic load orientation with deformation and, consequently, it alters the magnitude of the loads due to different deformed shapes. This work does not consider the impact of linearizing the local effective angle of attack, which was investigated in Ref. [29] for static solutions.

Because UM/NAST always computes the aerodynamic loads in the deformed shape based on the local effective angle of attack in Eq. (6), the “fully linear” case (linear kinematics and nonfollower aerodynamics) of this work does not exactly reproduce a standard linear aeroelastic analysis of off-the-shelf solvers such as MSC Nastran SOL 144 [34]. However, Sec. 4.1 shows that the differences between a true linear static aeroelastic solution and the fully linear case considered here are slight. In all cases, the local forces and moments per unit length in Eq. (5) or (8) are integrated along the span using the undeformed beam element lengths, which prevents nonphysical load magnification due to structural stretching in the case of linear kinematics.

3. Wing Benchmark Case

The benchmark case considered in this work is the Pazy wing [12], an experimental model for geometrically nonlinear aeroelastic studies at very large deflections in low-speed flow. Section 3.1 summarizes the main features of the wing; Secs. 3.2 and 3.3 describe the structural and aerodynamic numerical models, respectively. The structural model used in this work was verified and validated in Ref. [19] and has not been modified. One of the two aerodynamic models considered in this study, which builds on potential flow unsteady thin airfoil theory with tip loss corrections, was also established in Ref. [19]. To further investigate the impact of wingtip effects, this study introduces a second aerodynamic model that replaces the tip loss correction with non-uniform aerodynamic coefficient derivatives obtained from a panel method analysis. Details on the Pazy wing experimental model and data from structural and aeroelastic tests are available upon request through the

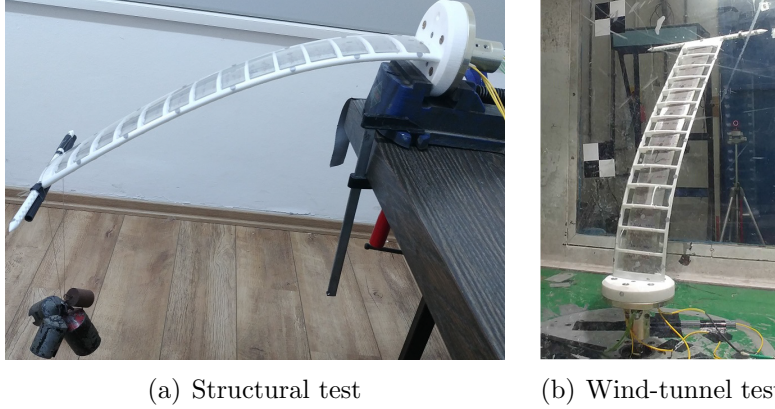


Figure 1: Pazy wing benchmark case (reprinted from Ref. [12], copyright 2022 by Yaron Ben-Shmuel and Moshe Tur).

AePW3-LDWG¹. The numerical model properties and results from this work are publicly available for download from the authors’ GitHub repository².

3.1. Experimental Model

Figure 1 shows the Pazy wing subject to structural [Fig. 1(a)] and aerodynamic [Fig. 1(b)] loading. The wing was developed at the Technion as part of the AePW3-LDWG activities and has a semispan $\ell = 0.55$ m, a chord $c = 0.1$ m, and a uniform untwisted NACA 0018 airfoil. The structure consists of a 7075 aluminum spar, a 3D-printed nylon 12 frame, an Orallight polyester skin providing the aerodynamic shape, and a 3D-printed tip rod to attach masses for altering the wing dynamics. Reference [12] details the experimental model design, ground vibration testing, and the first round of wind-tunnel tests focused on the wing static aeroelastic response and hard flutter crossing mechanism. Reference [13] discusses the second round of wind-tunnel tests that explored the hump mode flutter mechanism.

3.2. Low-Order Structural Model

The beam structural model was derived from the built-up finite element model (FEM) in Fig. 2 for the studies in Ref. [19] using an in-house structural

¹<https://nescacademy.nasa.gov/workshops/AePW3/public/wg/largedeflection>

²<https://github.com/UM-A2SRL/AePW3-LDWG.git>

model order reduction framework, the University of Michigan’s Enhanced FEM2Stick (UM/EF2S) [35]. The reference axis of the beam model is located at 44% chord in the airfoil symmetry plane and is discretized in 15 strain-based beam elements. This corresponds to a beam model size of 60 strain DOF (four strain DOF per beam element) compared with the original built-up FEM size of 41,946 displacement DOF (estimated assuming six displacement DOF times 6,991 built-up FEM nodes), for a model order reduction factor of 700. The inertia properties of the beam model are described by rigid-body elements at the beam element ends. These rigid-body elements are defined by their masses, mass offsets from the beam reference axis, and inertia tensors about the local centers of mass, which are computed based on data from the built-up FEM mass matrix. The stiffness properties consist of the elements of the cross-sectional stiffness matrix of the strain-based formulation of Ref. [30]. The strain-based stiffness matrix for each beam element has the axial, torsional, out-of-plane bending, and in-plane bending stiffness constants on the diagonal, and generally non-zero off-diagonal terms describing coupling effects between strain measures. These quantities are identified based on six linearly independent static solutions of the built-up FEM for small deflections about a reference shape, such as the undeformed shape, and are constant with structural deflections due to the assumed linear stress-strain constitutive law (no material nonlinearity). As mentioned in Sec. 2.1, the strain-based formulation of Ref. [30] captures geometrically nonlinear without requiring stiffness matrix updates due to the choice of strains as the independent structural DOF, differently from displacement-based formulations. While the stiffness matrix does not require updates, every other term in the equilibrium equation, Eq. (2), requires to be updated at every time step. In dynamic aeroelastic calculations, the generalized mass and damping matrices and two of the inflow matrices in Eq. (4) must also be updated for the current deformed shape.

The beam model was assessed in Ref. [19] by comparing modal and static results with solutions for the built-up FEM and experimental data from ground vibration testing (GVT) and static bending and torsion tests [19]. Modal and static solutions for the built-up FEM were obtained using the MSC Nastran modal solver SOL 103 [36] and the nonlinear structural solver SOL 400 [37], respectively. The comparisons showed that the beam model captures the first five natural frequencies of the Pazy wing with the same accuracy as the built-up FEM (errors below 1%, see Table 1) while requiring up to 700 times fewer DOF. The model also predicts the same static deformed

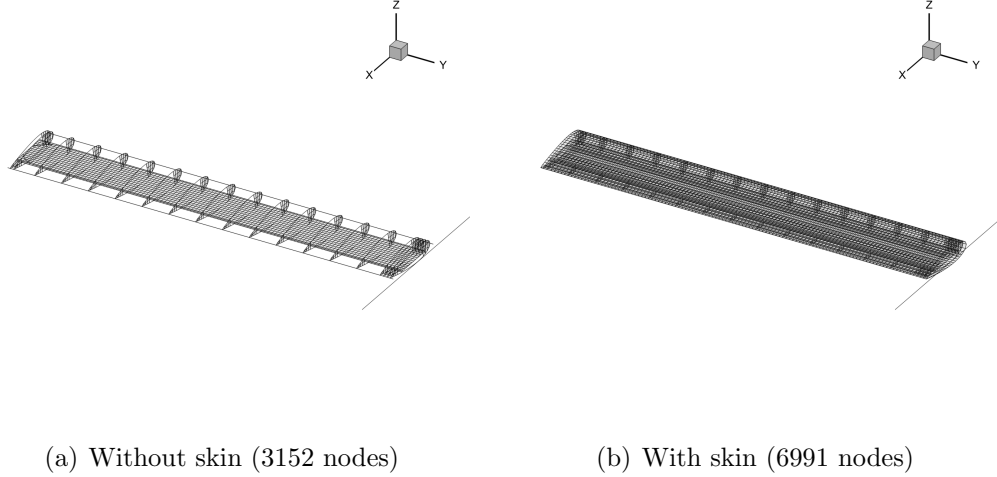


Figure 2: Pazy wing built-up FEM [19].

shapes as the built-up FEM in the 0–45% semispan tip deflection range [19]. Beyond requiring much fewer DOF, the beam model is more numerically stable than the built-up FEM, which fails in nonlinear solutions in the version with skin.

3.3. Low-Order Aerodynamic Model

The Pazy wing low-order aeroelastic model couples the beam model described in Sec. 3.2 with the potential flow unsteady thin airfoil theory of Peters et al. [31]. The low-order aerodynamic model consists of 15 strips each

Table 1: Natural frequencies in undeformed shape (with skin).

GVT [12]		SOL 103 [19]			UM/NAST [19]	
#	Mode	Freq. (Hz)	Freq. (Hz)	Δ_{Test} (%)	Freq. (Hz)	Δ_{FEM} (%)
1	OOP1	4.26	4.19	−1.58	4.19	−0.04
2	OOP2	28.50	28.49	−0.02	28.49	0.00
3	T1	42.00	41.97	−0.08	41.88	−0.21
4	IP1	60.70	82.92	1.74	83.06	0.17
5	OOP3	81.50	104.98	72.95	105.89	0.87

associated with a beam element for a total of 90 additional aerodynamic DOF in dynamic aeroelastic solutions (6 per strip). Two model variants are considered that implement user-specified cross-sectional aerodynamic coefficient derivatives but differ in the treatment of wingtip effects.

The first aerodynamic model, introduced in Ref. [19], features a uniform flat-plate airfoil and captures wingtip effects using tip loss corrections tuned on a set of static aeroelastic solution for different combinations of flow speed U and root angle of attack α_r . In this first model, the aerodynamic load distribution is corrected by the weighting factor

$$W(\bar{s}, U, \alpha_r) = 1 - \exp[-\tau(U, \alpha_r)(1 - \bar{s})] \quad (9)$$

that depends on the non-dimensional spanwise position $\bar{s} = s/\ell$ and on the tip loss factor

$$\tau(U, \alpha_r) = \tau_0(\alpha_r) + \tau_1(\alpha_r)U + \tau_2(\alpha_r)U^2 \quad (10)$$

The tip loss factor (10) is a function of the flow speed U and root angle of attack α_r . The coefficients τ_0 , τ_1 , and τ_2 are tuned such that tip vertical displacements from selected static aeroelastic solutions of the UM/NAST model match the ones computed by coupling the beam structure with the vortex-lattice method (VLM) in the UM/NAST-VLM model. The values of these coefficients resulting from the tuning process, conducted in Ref. [19], are reported in Table 2 for completeness. To highlight the order of magnitude of the tip loss correction introduced, Fig. 3 shows spanwise distributions of the weighting factor in Eq. (9) obtained using the coefficients in Table 2 at the root angle of attack $\alpha_r = 7$ deg for three representative flow speeds. The low-order aeroelastic model based on tip loss factors was verified and validated in [19] through static aeroelastic and flutter analyses. The model accurately captured static aeroelastic responses for a wide range of operating conditions, but predicted the hump mode flutter onset with errors up to 7.5%, with lower discrepancies at larger deflections.

To further investigate the sensitivity to wingtip effects, this work introduces a second aerodynamic model that also builds on the theory of Peters et al. [31] but replaces the tip loss factors with non-uniform cross-sectional aerodynamic coefficient derivatives. These aerodynamic coefficient derivatives are obtained from a rigid aerodynamic analysis of the Pazy wing undeformed shape at root angle of attack $\alpha_r = 1$ deg [14] using the ZAERO panel method solver [38]. This second low-order aerodynamic model does not require higher-order solutions for tuning the tip loss corrections, making it more

Table 2: Tip loss factor coefficients for Eq. (10).

α_r (deg)	τ_0 (-)	τ_1 (s/m)	τ_2 (s ² /m ²)
0	6.58	—	—
1	6.29	3.33×10^{-2}	-5.56×10^{-4}
2	6.00	5.19×10^{-2}	-8.59×10^{-4}
3	5.92	6.01×10^{-2}	-1.06×10^{-3}
4	5.91	6.49×10^{-2}	-1.23×10^{-3}
5	6.08	5.98×10^{-2}	-1.28×10^{-3}
6	6.43	4.43×10^{-2}	-1.19×10^{-3}
7	6.88	2.30×10^{-2}	-1.02×10^{-3}

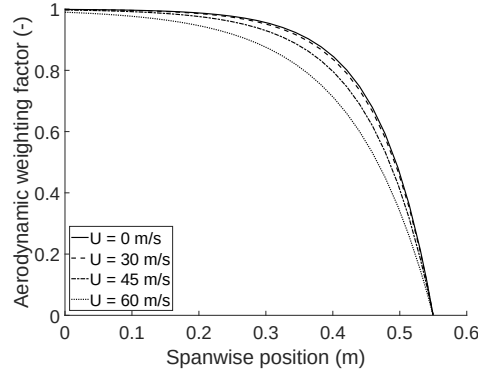


Figure 3: Aerodynamic weighting factor at $\alpha_r = 7$ deg and $U = 0, 30, 45, 60$ m/s.

suitable for studying the impact of different combinations of geometrically nonlinear effects. Thus, UM/NAST results based on the two low-order aerodynamic models are first compared with each other to understand how the modeling of wingtip effects impacts aeroelastic predictions. Next, only the UM/NAST model based on the second low-order aerodynamic model with non-uniform aerodynamic coefficient derivatives is considered to explore the analyze the impact of different geometrically nonlinear effects on the results. The UM/NAST low-order aeroelastic model based on the first aerodynamic model is hereafter referred to as the UM/NAST model based on tip loss factors; the low-order aeroelastic model based on the second aerodynamic model is hereafter referred to as the current UM/NAST model.

The non-uniform aerodynamic properties provided in Ref. [14] consist of the lift curve slope C_{l_α} and the non-dimensional center of pressure position along the chord r_{cp} measured from the leading edge. The quarter-chord pitching aerodynamic moment coefficient curve slope for UM/NAST is obtained as

$$C_{m_\alpha} = -C_{l_\alpha} (r_{cp} - 0.25) \quad (11)$$

Figure 4 shows the spanwise distributions of C_{l_α} and C_{m_α} used in this work (given in the local zero-lift aerodynamic frame for each beam node) along with the non-dimensional center of pressure position r_{cp} . The spanwise variation in C_{l_α} and r_{cp} is a consequence of three-dimensional aerodynamic effects and causes C_{m_α} to also vary along the span according to Eq. (11). The values of C_{l_α} and C_{m_α} are set to zero at the wing tip in Fig. 4 and in the analyses.

4. Results

The results discussed in this section highlight how geometrically nonlinear effects impact wing aeroelastic behaviors at large deflections, using the Pazy wing as the benchmark case. The study considers the Pazy wing with skin unless stated otherwise, and neglects the wing self-weight. Tip displacements are recovered at the half-chord point and tip twist angles are estimated from the differential displacements of the tip leading- and trailing-edge points. The static and modal results consider a 0.01 kg mass on the tip rod 0.01 m behind the trailing edge, consistently with the wind-tunnel tests in Ref. [12]. These results highlight how different geometrically nonlinear effects impact static deformed shapes and the associated natural frequencies, which eases interpreting the flutter and post-flutter results. Next, a flutter analysis of the

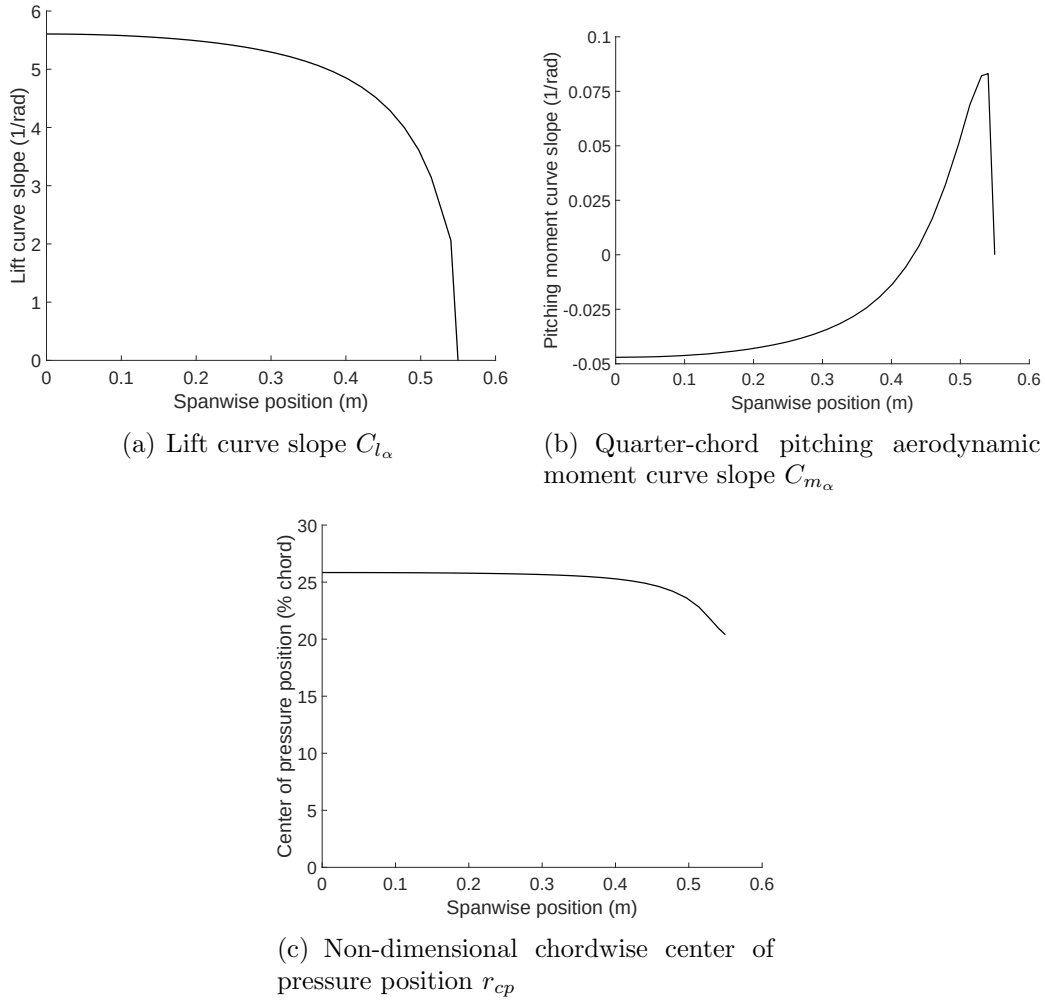


Figure 4: Aerodynamic coefficient derivatives and non-dimensional center of pressure position along the span.

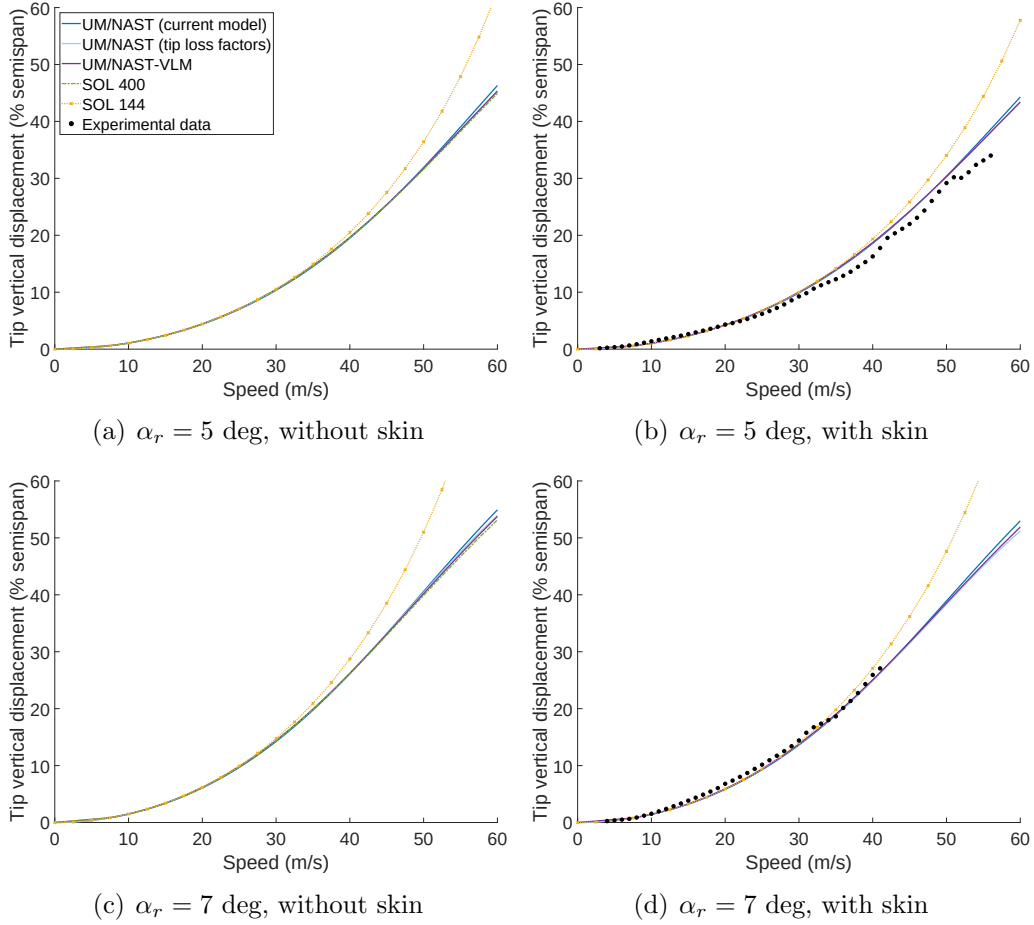


Figure 5: Tip vertical displacements for $U = 0 \rightarrow 60$ m/s at $\alpha_r = 5, 7$ deg.

undeformed wing is conducted to highlight the impact of three-dimensional aerodynamic effects on the results. Finally, flutter and limit-cycle oscillation results in deformed shape without the tip mass are presented to investigate the hump mode flutter mechanism at large deflections [13].

4.1. Static Aeroelastic Response

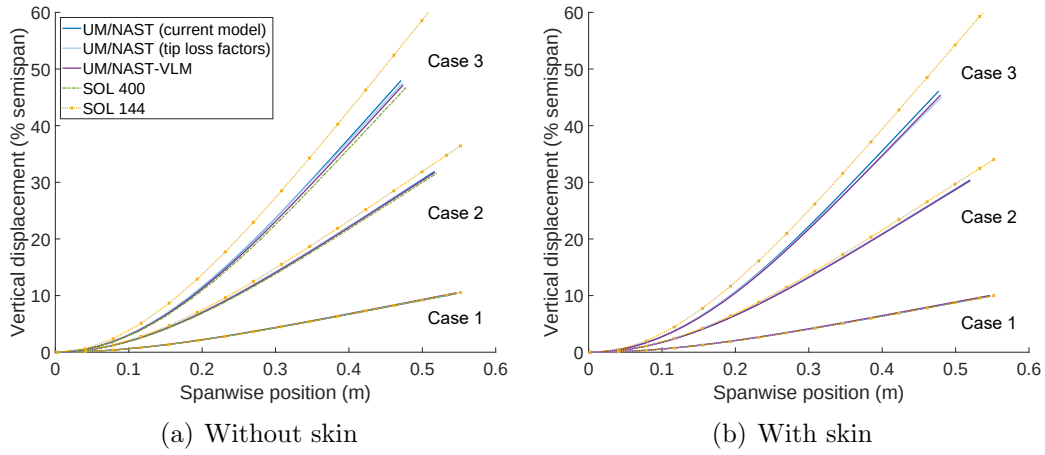


Figure 6: Selected deformed shapes.

Table 3: Selected tip vertical displacements (% semispan).

(a) $\alpha_r = 5$ deg, without skin						
	SOL 400	UM/NAST-VLM	Δ_{FEM} (%)	UM/NAST	Δ_{FEM} (%)	Δ_{VLM} (%)
30	10.47	10.48	0.09	10.34	−1.20	−1.29
40	19.59	19.63	0.24	19.49	−0.47	−0.71
50	31.59	31.76	0.53	31.92	1.04	0.51
(b) $\alpha_r = 7$ deg, without skin						
	SOL 400	UM/NAST-VLM	Δ_{FEM} (%)	UM/NAST	Δ_{FEM} (%)	Δ_{VLM} (%)
30	14.38	14.40	0.14	14.23	−1.04	−1.18
40	26.11	26.21	0.39	26.12	0.05	−0.33
50	39.88	40.19	0.76	40.60	1.80	1.03
(c) $\alpha_r = 5$ deg, with skin						
	Test [12]	UM/NAST-VLM	Δ_{Test} (%)	UM/NAST	Δ_{Test} (%)	Δ_{VLM} (%)
30	9.19	10.00	8.84	9.87	7.41	−1.31
40	16.15	18.72	15.90	18.57	15.02	−0.76
50	29.03	30.29	4.35	30.41	4.75	0.39
(d) $\alpha_r = 7$ deg, with skin						
	Test [12]	UM/NAST-VLM	Δ_{Test} (%)	UM/NAST	Δ_{Test} (%)	Δ_{VLM} (%)
30	14.41	13.77	−4.45	13.60	−5.61	−1.21
40	25.93	25.07	−3.32	24.96	−3.71	−0.41
50	n.a.	38.57	n.a.	38.92	n.a.	0.92

The static aeroelastic discussion starts by assessing the current UM/NAST model against the UM/NAST model based on tip loss factors. The comparison also includes numerical results from the UM/NAST-VLM model, the built-up FEM analyzed in the MSC Nastran nonlinear structural solver SOL 400 [37] subject to frozen UM/NAST-VLM converged aerodynamic loads, and the built-up FEM analyzed in the MSC Nastran linear static aeroelastic solver SOL 144 [34] coupled with the linear VLM. Reference experimental data are from the Technion wind-tunnel tests [12]. Results are presented for both the models with and without skin for varying flow speed $U = 0 \rightarrow 60$ m/s at root angle of attack $\alpha_r = 5, 7$ deg. Figure 5 and Table 3 present tip vertical displacements while Fig. 6 shows deformed shapes for: $\alpha_r = 5$ deg and $U = 30$ m/s (Case 1, small deflections); $\alpha_r = 5$ deg and $U = 50$ m/s (Case 2, large deflections); and $\alpha_r = 7$ deg and $U = 55$ m/s (Case 3, very large deflections). Table 3 compares UM/NAST results from the current model with numerical solutions from SOL 400 and UM/NAST-VLM and experimental data, showing the relative differences Δ_{FEM} , Δ_{VLM} , and Δ_{Test} .

The current UM/NAST model predicts static aeroelastic responses with accuracy comparable to the UM/NAST model based on tip loss factors. Slight differences are only observable for the highest deflections and are explained considering that the tip loss factors in the UM/NAST model of Ref. [19], given by Eq. (10), vary with the operating condition while the non-uniform cross-sectional aerodynamic properties in the current UM/NAST model come from a panel method analysis of the wing undeformed shape [14]. However, the current model captures the UM/NAST-VLM tip vertical deflections within 1.3% with no need for higher-order solutions for tip loss factor tuning. Using aerodynamic properties based on a panel method analysis also allows for a physics-based modeling of wingtip effects, in contrast with tip loss factors. Predictions by the current UM/NAST model differ from the experimental data by up to 15% (see Table 3). These differences are attributed to viscosity, thickness, and skin wrinkling effects that are not modeled. However, the experimental data do not have error bounds and present fluctuations that may affect the comparison, particularly at $\alpha_r = 5$ deg. Overall, these results verify that the cross-sectional aerodynamic properties in Fig. 4 are correctly implemented into the current UM/NAST model. The results also reinforce that strip theory accurately captures static responses of very flexible wings up to extreme deflections, provided wingtip effects are considered. The results are relatively insensitive to the approach used to account for these effects.

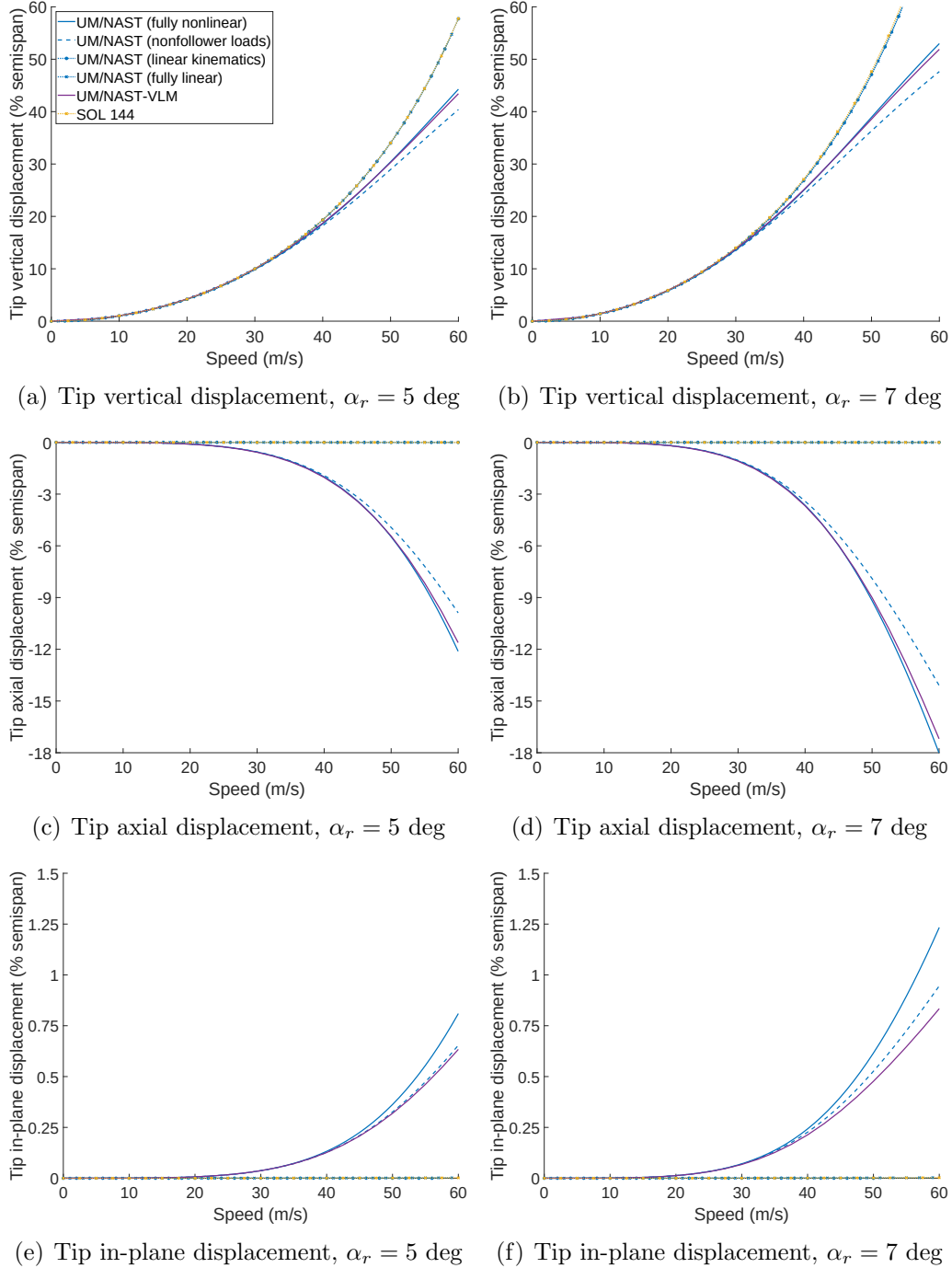


Figure 7: Tip displacements for different geometrically nonlinear effects.

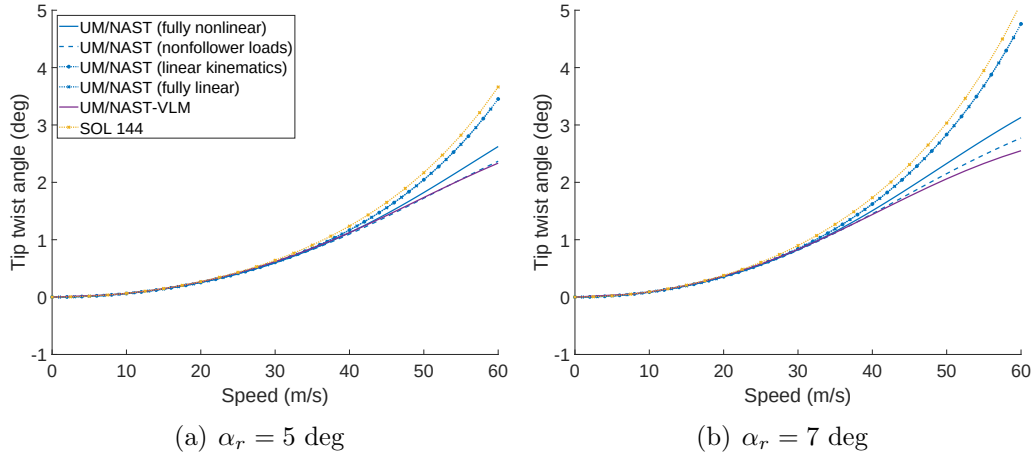


Figure 8: Tip twist angles for different geometrically nonlinear effects.

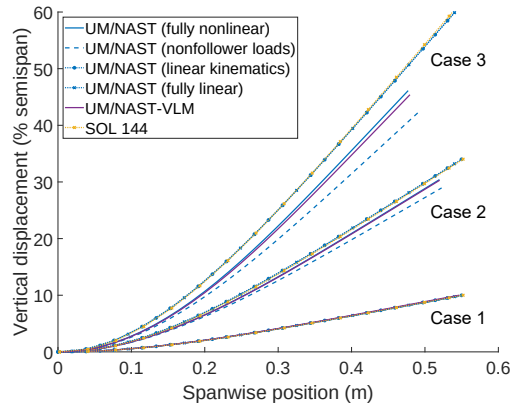


Figure 9: Selected deformed shapes for different geometrically nonlinear effects.

Table 4: Selected tip vertical displacements (% semispan) for different geometrically nonlinear effects.

(a) $\alpha_r = 5$ deg

U (m/s)	Fully nonlinear	Nonfollower loads	Δ (%)	Linear kinematics	Δ (%)	Fully linear	Δ (%)
30	9.87	9.82	−0.53	9.98	1.08	9.98	1.07
40	18.57	18.22	−1.90	19.32	3.99	19.31	3.98
50	30.41	28.94	−4.82	34.00	11.80	34.00	11.79
60	44.28	40.39	−8.78	57.72	30.34	57.71	30.33

(b) $\alpha_r = 7$ deg

U (m/s)	Fully nonlinear	Nonfollower loads	Δ (%)	Linear kinematics	Δ (%)	Fully linear	Δ (%)
30	13.60	13.47	−0.98	13.88	2.06	13.88	2.05
40	24.96	24.18	−3.16	26.82	7.45	26.82	7.43
50	38.92	36.32	−6.69	47.08	20.96	47.08	20.95
60	52.98	47.65	−10.07	79.52	50.08	79.51	50.07

Table 5: Selected tip twist angles (deg) for different geometrically nonlinear effects.

(a) $\alpha_r = 5$ deg

U (m/s)	Fully nonlinear	Nonfollower loads	Δ (%)	Linear kinematics	Δ (%)	Fully linear	Δ (%)
30	0.60	0.60	−0.61	0.61	1.11	0.61	1.10
40	1.12	1.10	−2.15	1.17	4.12	1.17	4.11
50	1.82	1.72	−5.40	2.05	12.23	2.05	12.22
60	2.62	2.37	−9.77	3.45	31.55	3.45	31.53

(b) $\alpha_r = 7$ deg

U (m/s)	Fully nonlinear	Nonfollower loads	Δ (%)	Linear kinematics	Δ (%)	Fully linear	Δ (%)
30	0.82	0.82	−1.12	0.84	2.13	0.84	2.12
40	1.51	1.45	−3.59	1.62	7.69	1.62	7.68
50	2.33	2.15	−7.58	2.83	21.72	2.83	21.71
60	3.13	2.77	−11.44	4.76	52.06	4.76	52.05

Next, the current UM/NAST model with skin is used to explore the role of geometrically nonlinear effects (trends without skin are the same). Figures 7 and 8 show the tip displacements and twist angles for varying flow speed; Fig. 9 compares deformed shapes for the conditions in Fig. 6. The blue solid lines in Figs. 7 to 9 indicate the fully nonlinear UM/NAST results (nonlinear kinematics and follower aerodynamics); the blue dashed lines and the blue dotted lines with circles show the results obtained with nonlinear kinematics and nonfollower aerodynamics or linear kinematics and follower aerodynamics, respectively; and the blue dotted lines with crosses indicate the fully linear results (linear kinematics and nonfollower aerodynamics). To highlight the role of three-dimensional aerodynamic effects, the figures also show the fully nonlinear UM/NAST-VLM solutions (purple solid lines) and the SOL 144 solutions [34] (yellow dotted lines with crosses).

Follower effects play a role as tip deflections exceed 10–15% semispan. Neglecting these effects while keeping nonlinear kinematics (blue dashed lines) gives smaller tip vertical displacements than the fully nonlinear case: nonfollower aerodynamic forces do not remain normal to the beam reference axis as the wing deflects, producing an axial component that does not contribute to out-of-plane bending. Smaller tip vertical displacements translate to smaller (in magnitude) axial displacements due to weaker shortening effects. The impact on in-plane displacements and twist angles is slighter.

Nonlinear kinematics also plays a role as tip deflections exceed 10–15% semispan. Assuming linear kinematics with follower aerodynamics (blue dotted lines with circles) severely overestimates tip vertical displacements and twist angles while it misses other displacement components. The deformed shapes based on linear kinematics are highly stretched and depart from those based on nonlinear kinematics at larger deflections (Case 2 and 3 in Fig. 9). With linear kinematics, follower aerodynamics should give smaller tip vertical displacements for a given condition because aerodynamic loads do not remain normal to the *undeformed* beam reference axis as the wing deflects (linearly). However, the results with and without follower aerodynamics are the same in the case of linear kinematics.

Figures 7 to 9 show that the UM/NAST fully linear results lie on top of those from SOL 144. Thus, not only the beam model accurately represents the built-up FEM [19], but also the low-order aerodynamic model captures steady wingtip effects. These results corroborate the findings for the fully nonlinear case (Figs. 5 and 6) and verify the UM/NAST linear static aeroelastic capabilities. The agreement between the UM/NAST and SOL 144 fully

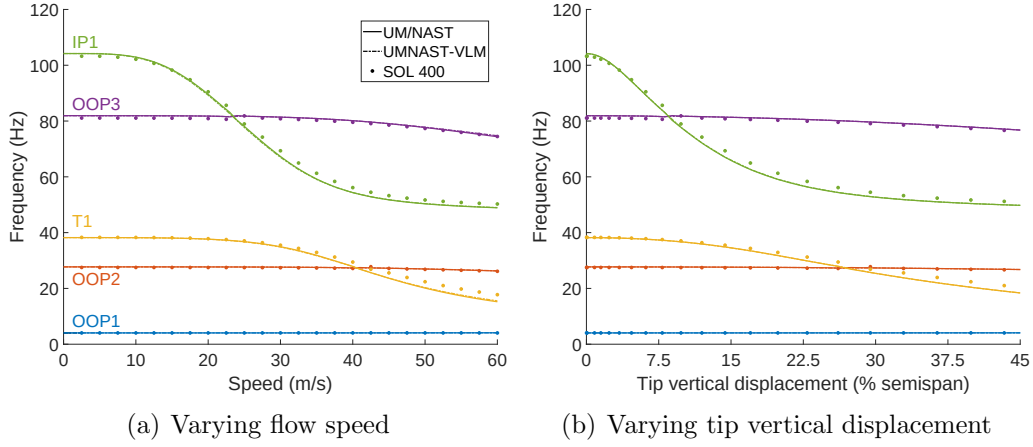


Figure 10: Natural frequency verification at $\alpha_r = 7^\circ$.

linear solutions shows that computing the aerodynamic loads in deformed shape based on the geometrically nonlinear local effective angle of attack has only a slight impact on the results. Differences between the fully nonlinear UM/NAST and UM/NAST-VLM results are small but increase with the flow speed. This is explained by the stronger wingtip effects at larger deflections, which are associated with a shorter effective wing semispan. The UM/NAST-VLM model captures this effect by computing the aerodynamic loads on the deformed VLM geometry; the UM/NAST model does not because it uses aerodynamic properties computed in undeformed shape. Thus, UM/NAST-VLM predicts smaller deflections than UM/NAST at the highest flow speeds, though the impact of this aerodynamic geometrical nonlinearity is small, as also observed in Ref. [28].

Tables 4 and 5 compare UM/NAST tip vertical displacements and twist angles at selected conditions. The quantity Δ refers to the percentage differences with respect to the fully nonlinear results. Nonlinear kinematics dominates the static aeroelastic response: neglecting this effect overestimates tip vertical displacements and twist angles by up to about 50% while neglecting follower aerodynamics underestimates them by about 10%. There is no difference between follower or nonfollower aerodynamics for linear kinematics.

4.2. Natural Frequencies in Deformed Shape

This section investigates the impact of geometrically nonlinear effects on the natural frequencies in deformed shape. Before that, it is necessary to

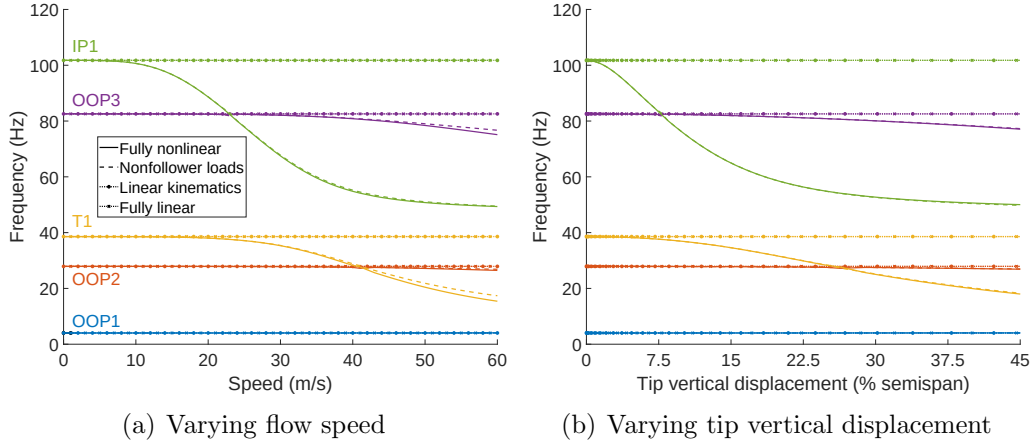


Figure 11: Natural frequencies at $\alpha_r = 7$ deg for different geometrically nonlinear effects.

verify the UM/NAST fully nonlinear modal analysis. In this paper, the term “fully nonlinear modal analysis” refers to a local modal analysis conducted by linearizing the structural equations of motion in Sec. 2 about the static (geometrically nonlinear) deformed shape at a given operating condition (flow speed and root angle of attack). Because the natural frequencies and the corresponding mode shapes in deformed shape result from a linearized analysis, they have similar meaning as the corresponding quantities evaluated in undeformed shape, except for the impact of static geometrically nonlinear effects. These quantities are only used to gain additional insight into the effects of the large static deflections but not used to solve the system of equations as one would do with modal projection in the linear sense or with nonlinear normal modes in a nonlinear sense (*e.g.*, [39, 40]). In UM/NAST, the natural frequencies for a given static deformed shape are computed by solving the generalized eigenvalue problem associated with the strain-based stiffness matrix, which is constant with deformation, and the corresponding linearized mass matrix associated with the flexible (strain) DOF evaluated for that deformed shape. The resulting natural frequencies are verified the ones for the built-up FEM obtained from SOL 400 [37]. Each SOL 400 solution computes the static deformed shape of the built-up FEM subject to frozen UM/NAST-VLM converged aerodynamic loads for given flow speed and root angle of attack, followed by a local linearized modal analysis. Figure 10 compares the results from UM/NAST, UM/NAST-VLM, and SOL 400 at root angle of attack $\alpha_r = 7$ deg. The results are for the case without

skin because the built-up FEM with skin fails in SOL 400 [19]. Natural frequencies associated with different structural modes are distinguished by their colors and labeled according to the corresponding mode type as identified in the undeformed shape (zero flow speed). The solid lines are the results from the current UM/NAST model; the dash-dotted lines are the results from UM/NAST-VLM; the circles are the results from SOL 400.

The UM/NAST and UM/NAST-VLM results overlap, showing that approximating wingtip effects using strip theory does not impact the modal scenario, as expected from the agreement between the static deformed shapes (Sec. 4.1). The UM/NAST fully nonlinear model captures the trend of the natural frequencies of the built-up FEM despite using much fewer DOF. Tables 6 and 7 compare natural frequencies for selected conditions. The OOP1 frequency is practically constant with the flow speed. Differences between the beam model and built-up FEM predictions increase slightly from 0.1% to 1% with the flow speed due to the minor differences in the static deformed shapes (Sec. 4.1). The OOP2 and OOP3 frequencies also vary only slightly with the flow speed and the results from the beam model are within less than 1% of the ones from the built-up FEM. The T1 frequency reduces by more than 50% as the flow speed increases. The beam model underestimates the built-up FEM frequency by up to 15%, though absolute differences are below 2.5 Hz. The beam model also predicts a larger IP1 frequency drop, though results are within 3% of the built-up FEM. Next, the discussion shifts to the role of geometrically nonlinear effects. The results are for the current UM/NAST model with skin considered in the flutter analyses (trends with and without skin are similar). Figure 11 shows the first five natural frequencies for different geometrically nonlinear effects. Linear kinematics completely misses any frequency variation with the flow speed. In the case of nonlinear kinematics, considering nonfollower aerodynamics does not impact the OOP1 frequency and impacts the OOP2, OOP3, and IP1 frequencies slightly. The most appreciable impacts are on the T1 frequency at flow speeds beyond 40 m/s, where the results with and without follower aerodynamics differ by up to about 13%. The frequencies with follower aerodynamics show a higher decrease with the flow speed due to larger wing deflections (Sec. 4.1). However, the results with follower and nonfollower aerodynamics overlap when the frequencies are plotted as a function of the tip vertical displacement in Fig. 11(b), with the flow speed as an implicit parameter of the curves. This shows that the modal scenario of a very flexible wing at a given root angle of attack only depends on its deformed shape regardless of the flow speed that

produces such shape [4].

To summarize, neglecting nonlinear kinematics misses the changes in the modal scenario with deformation. Neglecting follower aerodynamics misses the correct map from flow speed and root angle of attack to natural frequencies due to underestimated static deflections. However, the results with and without follower aerodynamics are the same once natural frequencies are plotted as a function of the tip vertical displacement, showing that follower aerodynamics effects does not change the qualitative trends in the modal scenario. These effects mostly impact the frequency of the torsion-dominated mode at the highest flow speeds, while the analyzed bending-dominated frequencies vary only slightly. This can be explained by the high sensitivity of torsion-dominated modes to static deflections that couple torsion and in-plane bending motions.

Table 6: Natural frequency verification at $\alpha_r = 7$ deg for $U = 10, 20, 30$ m/s.

(a) $U = 10$ m/s, tip vertical displacement = 1.50% semispan

		SOL 400	UM/NAST	
#	Mode type	Freq. (Hz)	Freq. (Hz)	Δ (%)
1	OOP1	4.04	4.05	0.17
2	OOP2	27.52	27.72	0.73
3	T1	38.27	38.18	-0.23
4	OOP3	81.07	81.90	1.02
5	IP1	102.15	102.92	0.75

(b) $U = 20$ m/s, tip vertical displacement = 6.16% semispan

		SOL 400	UM/NAST	
#	Mode type	Freq. (Hz)	Freq. (Hz)	Δ (%)
1	OOP1	4.04	4.05	0.16
2	OOP2	27.50	27.70	0.74
3	T1	37.82	37.66	-0.42
4	OOP3	80.88	81.77	1.11
5	IP1	90.49	89.59	-0.99

(c) $U = 30$ m/s, tip vertical displacement = 14.38% semispan

		SOL 400	UM/NAST	
#	Mode type	Freq. (Hz)	Freq. (Hz)	Δ (%)
1	OOP1	4.05	4.05	0.14
2	OOP2	27.42	27.63	0.75
3	T1	35.50	34.94	-1.58
4	IP1	69.34	67.30	-2.94
5	OOP3	80.81	81.40	0.73

Table 7: Natural frequency verification at $\alpha_r = 7$ deg for $U = 40, 50, 60$ m/s.

(a) $U = 40$ m/s, tip vertical displacement = 26.11% semispan

		SOL 400	UM/NAST	
#	Mode type	Freq. (Hz)	Freq. (Hz)	Δ (%)
1	OOP1	4.06	4.06	0.08
2	OOP2	27.15	27.30	0.54
3	T1	29.46	27.90	-5.27
4	IP1	56.14	54.36	-3.17
5	OOP3	79.56	80.13	0.72

(b) $U = 50$ m/s, tip vertical displacement = 39.88% semispan

		SOL 400	UM/NAST	
#	Mode type	Freq. (Hz)	Freq. (Hz)	Δ (%)
1	OOP1	4.05	4.07	0.46
2	T1	22.39	20.11	-10.19
3	OOP2	26.83	26.96	0.48
4	IP1	51.70	50.32	-2.66
5	OOP3	77.34	77.67	0.43

(c) $U = 60$ m/s, tip vertical displacement = 53.15% semispan

		SOL 400	UM/NAST	
#	Mode type	Freq. (Hz)	Freq. (Hz)	Δ (%)
1	OOP1	4.04	4.09	1.38
2	T1	17.76	15.21	-14.33
3	OOP2	26.14	26.25	0.42
4	IP1	50.26	48.90	-2.71
5	OOP3	74.47	74.38	-0.12

Table 8: Flutter points in undeformed shape.

(a) Without tip mass (hump mode mechanism)

UM/NAST						
Quantity	Unit	SOL 145	Ref. [19]	Δ_{FEM} (%)	Current	Δ_{FEM} (%)
Flutter speed	m/s	67.30	72.40	7.58	87.51	30.03
Flutter frequency	Hz	34.72	34.72	0.00	32.15	−7.40
Divergence speed	m/s	100.97	98.32	−2.62	99.51	−1.44

(b) With tip mass (hard flutter crossing mechanism)

UM/NAST						
Quantity	Unit	SOL 145	Ref. [19]	Δ_{FEM} (%)	Current	Δ_{FEM} (%)
Flutter speed	m/s	86.53	91.77	6.06	91.79	6.08
Flutter frequency	Hz	16.16	12.80	−20.76	13.65	−15.50
Divergence speed	m/s	100.97	98.31	−2.63	99.51	−1.45

4.3. Flutter Onset in Undeformed Shape

This section analyzes how the modeling of three-dimensional aerodynamic effects impact the flutter onset in undeformed shape. This study serves to both assess the current UM/NAST model for a dynamic case and to ease explaining the impact of geometrically nonlinear effects on the flutter boundary in deformed shape in Sec. 4.4. The flutter results are for the Pazy wing with skin, with and without the tip mass. Without the tip mass, the critical flutter mechanism is a hump mode; with the tip mass, the critical flutter mechanism is a hard flutter crossing at higher flow speed and lower frequency [12]. Table 8 compares the UM/NAST results based on the current model and on tip loss factors with the results from the MSC Nastran linear flutter analysis solver SOL 145 [34], obtained by coupling the built-up FEM with the doublet-lattice method (DLM). The UM/NAST flutter analysis consists of p -method aeroelastic eigenvalue analyses at varying flow speed $U = 0 \rightarrow 120$ m/s and zero root angle of attack $\alpha_r = 0$ deg without gravity. Because the UM/NAST model does not deflect under these conditions, the flutter solution is akin to that implemented in SOL 145 except for the use of different analysis method (p -method in UM/NAST and p - k method in SOL 145, respectively).

The current UM/NAST model predicts the hard flutter crossing mechanism with the tip mass with the same accuracy as the UM/NAST model based on tip loss factors: the flutter speeds from the two models are both within 6% of SOL 145. The current UM/NAST model predicts the hard flutter crossing frequency within 15% of SOL 145, showing higher accuracy than the model based on tip loss factors. The current UM/NAST model also predicts the divergence speed within 1.4% of SOL 145, improving on the 2.6% error achieved using tip loss factors.

However, the current UM/NAST model predicts the hump mode flutter mechanism less accurately than the model based on tip loss factors. While that model predicts the hump mode flutter speed within 7.6% of SOL 145, and matches the SOL 145 flutter frequency, the current model gives a 30% higher flutter speed and a 7.4% lower flutter frequency than SOL 145. These differences are attributed to the high sensitivity of the hump mode flutter mechanism to three-dimensional unsteady aerodynamic effects. This high sensitivity arises from the nature of hump modes, whose instability point can shift significantly for small damping variations. As a result, even small differences in capturing the impact of three-dimensional unsteady aerodynamic effects on modal damping may significantly alter the predicted flutter

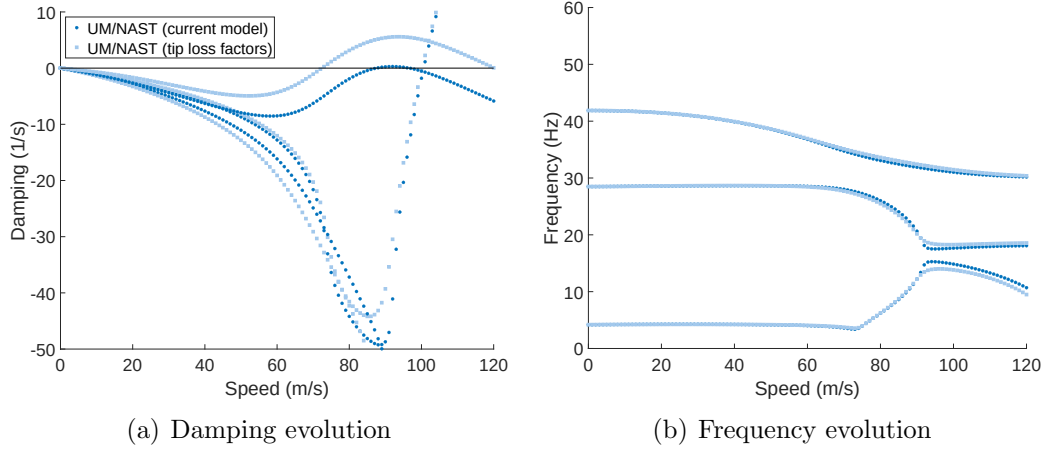


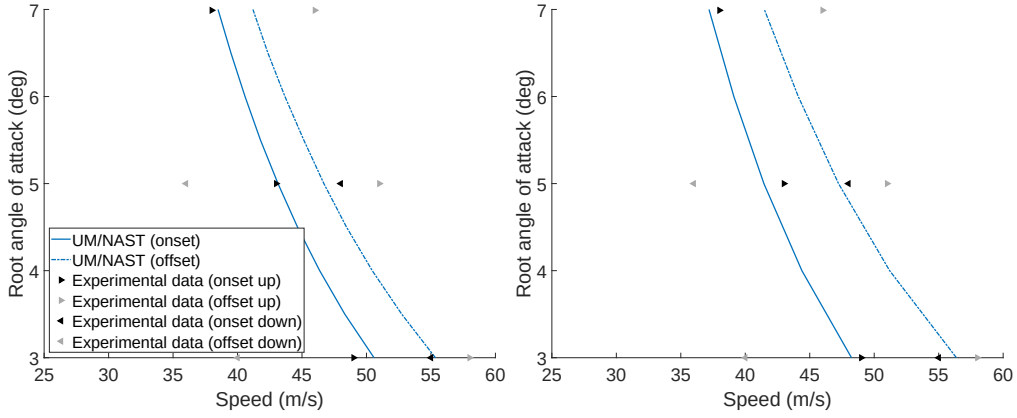
Figure 12: First three aeroelastic eigenvalues at $\alpha_r = 0$ deg (without tip mass).

speed and frequency. To highlight this point, Fig. 12 shows the aeroelastic eigenvalues associated with the first three structural frequencies. The hump mode is barely unstable for the current UM/NAST model while the hard flutter crossing predicted by the two UM/NAST models overlap. In fact, the hard flutter crossing is less influenced by three-dimensional unsteady aerodynamic effects, while the hump mode cannot be even captured by neglecting these effects (that is, considering two-dimensional aerodynamics), as shown in Ref. [19].

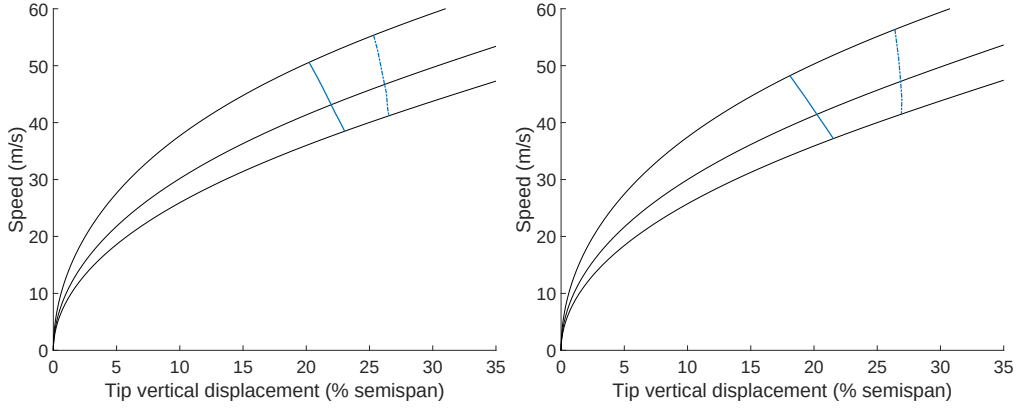
To summarize, predictions of hump mode flutter mechanisms in the linear regime are extremely sensitive to the approach for modeling three-dimensional unsteady aerodynamic effects. Different approaches may result in large discrepancies between the predicted flutter onset speeds and frequencies. Additionally, even the most accurate predictions based on strip theory present errors of about 8% compared with results based on higher-detailed unsteady aerodynamic modeling approaches such as the DLM. This finding contrast with the results in Sec. 4.1 showing that static aeroelastic predictions do not vary significantly with the approach to model three-dimensional steady aerodynamic effects.

4.4. Hump Mode Flutter Boundary in Deformed Shape

This section investigates the hump mode flutter boundary without tip mass for varying root angle of attack $\alpha_r = 3 \rightarrow 7$ deg. First, results from the

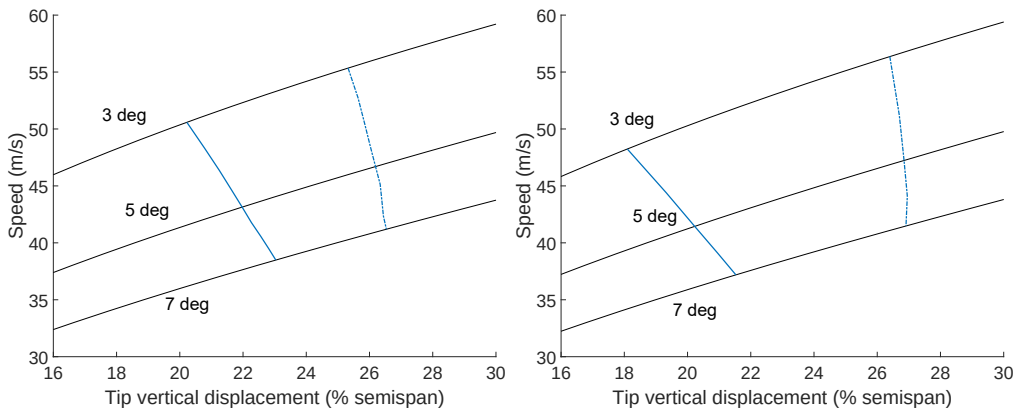


(a) Varying root angle of attack (Current) (b) Varying root angle of attack (Ref. [19])



(c) Varying deflection (Current)

(d) Varying deflection (Ref. [19])



(e) Zoom of Fig. 13(c)

(f) Zoom of Fig. 13(d)

Figure 13: Hump mode flutter boundary.

current UM/NAST model and from the UM/NAST model with tip loss factors are compared with experimental data [13] to show the impact of different approximations for wingtip effects. Next, the current UM/NAST model is used to explore the role of geometrically nonlinear effects. The UM/NAST results are obtained by linearizing the governing equations about the static deformed shape at each flow speed and root angle of attack, followed by a local p -method eigenvalue analysis [10]. The hump mode flutter onset for a given root angle of attack is determined as the lowest flow speed where a modal damping value becomes positive. The flutter offset is computed as the subsequent flow speed where that modal damping value becomes negative again. The experimental flutter boundary was characterized by means of increasing and decreasing flow speed sweeps at the root angles of attack $\alpha_r = 3, 5, 7$ deg. Due to hysteresis, the flutter offset from a flow speed sweep differs from the onset from the reversed sweep at the same root angle of attack, which highlights a subcritical bifurcation.

Figures 13(a) and 13(b) show the eigenvalue-based flutter onset and offset speeds for varying root angle of attack. Figures 13(c) and 13(d) show the results for varying tip vertical displacement, with the static equilibrium paths as dashed lines. Zooms of these results are given in Figs. 13(e) and 13(f). The current UM/NAST model and the one based on tip loss factors give closer flutter onsets at higher root angles of attack and larger deflections, but their predictions differ for smaller deflections. Specifically, the current UM/NAST model predicts a higher flutter onset speed and a lower offset speed, in line the results in the underformed shape (Sec. 4.3). However, the flutter boundaries from the two models show similar trends: the flutter onset speed decreases with increasing root angle of attack and the flutter offset speed also decreases but by a larger amount, such that the hump mode instability region shrinks.

The smaller differences between predictions from the two UM/NAST models at higher root angles of attack indicate a reduced influence of three-dimensional unsteady aerodynamic effects and a stronger impact of geometrically nonlinear effects. In other words, the coupling between different aeroelastic modes causing the instability is driven by three-dimensional unsteady aerodynamic effects at low deflections and by changes in the modal scenario in deformed shape at larger deflections. This suggests that accurate modeling of three-dimensional *unsteady* aerodynamics is critical to predict hump mode flutter mechanisms in the linear regime, while accurate modeling of three-dimensional *steady* aerodynamic effects (which impact the static deformed

shape) is essential to predict these mechanisms at larger deflections.

Table 9 compares the UM/NAST results with the experimental data for $\alpha_r = 3, 5$ and 7 deg, shown as markers in Figs. 13(a) and 13(b). The UM/NAST flutter onset and offset speeds are compared with the “onset up” data from the increasing flow speed sweep and the “onset down” data from the decreasing flow speed sweep, respectively. This ensures that numerical results from eigenvalue analyses are compared with consistent experimental data not affected by hysteresis. The current UM/NAST model captures the experimental flutter onset with higher accuracy than the model based on tip loss factors for $\alpha_r = 5, 7$ deg in the “onset up” case and for $\alpha_r = 3$ deg in the “onset down” case. While errors for the other cases increase by 1–2%, the experimental data are overall captured within 4%, like in the case of tip loss factors, though lowest and highest errors are achieved at different conditions. The experimental flutter offsets, shown as gray markers in Figs. 13(a) and 13(b), differ from the eigenvalue-based predictions by 2–20% (tip loss factors) and 4–26% (current model). This is because flutter offsets determined from eigenvalue analyses cannot capture experimental data affected by hysteresis [13]. For completeness, Fig. 14 shows the stability scenarios by the two UM/NAST models for $U = 5 \rightarrow 60$ m/s at $\alpha_r = 3, 5, 7$ deg.

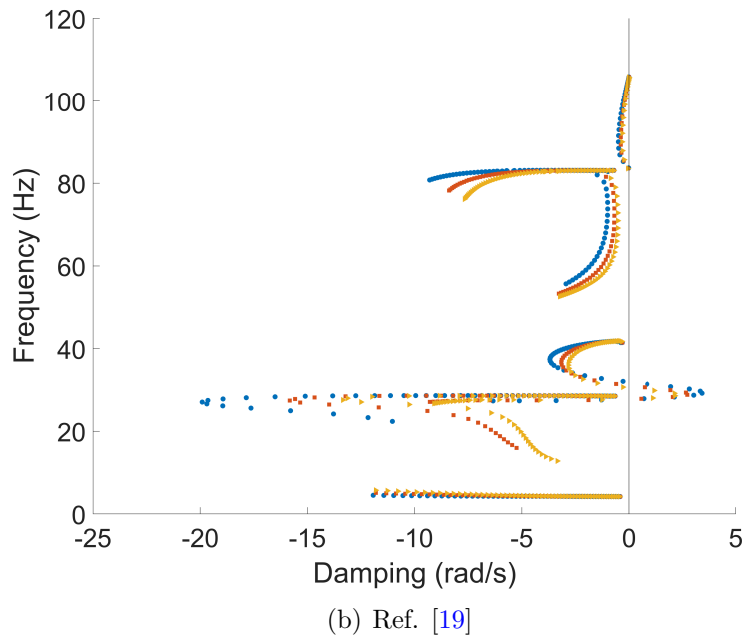
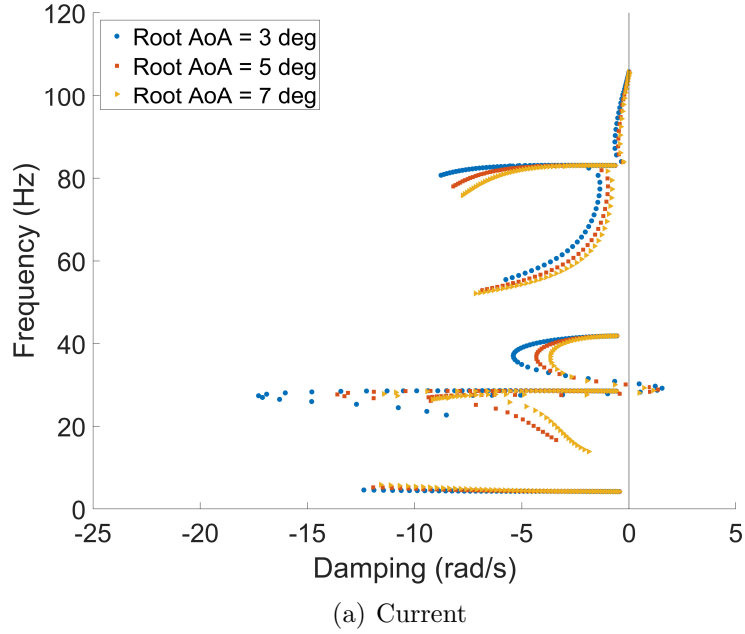


Figure 14: Root loci for $U = 5 \rightarrow 60$ m/s.

Table 9: Hump mode flutter onset speeds (m/s) for $\alpha_r = 3, 5, 7$ deg.

UM/NAST						
α_r (deg)	Flow speed sweep	Test [13]	Current	Δ_{Test} (%)	Ref. [19]	Δ_{Test} (%)
3	Increasing (onset up)	49.00	50.58	3.23	48.24	−1.55
5	Increasing (onset up)	43.00	43.14	0.32	41.44	−3.62
7	Increasing (onset up)	38.00	38.49	1.28	37.19	−2.14
3	Decreasing (onset down)	55.00	55.35	0.63	56.37	2.50
5	Decreasing (onset down)	48.00	46.70	−2.71	47.23	−1.60

Next, the current UM/NAST model is used to study the role of geometrically nonlinear effects. The discussion focuses on the results based on nonlinear kinematics with and without follower aerodynamics. The results based on linear kinematics are not shown because the hump mode flutter onset and offset speeds are close to the values in undeformed shape for low deflections, where the hump mode is barely unstable with the current low-order aerodynamic model (Sec. 4.3).

Figure 15 shows the hump mode flutter onset and offset speeds for varying root angle of attack and tip vertical displacement. The solid curves are the fully nonlinear results and the dashed curves are the results for nonfollower aerodynamics. The flutter onset and offset speeds are distinguished using different colors.

Neglecting follower aerodynamics predicts higher flutter onset and offset speeds at each root angle of attack. Table 10 compares the flutter speeds along with their percentage differences. The flutter onset speeds with and without follower aerodynamics differ by 1.3% to 1.8% as the root angle of attack increases while the offset speeds differ by 3.7% to 4%. These differences are explained considering that follower aerodynamics causes larger wing deflections and, consequently, larger changes in the frequency of the torsion-dominated mode with flow speed (which drive the hump mode flutter mechanism). Together, these effects shrink the hump mode instability range at each root angle of attack and shift it at lower flow speeds. Differences between the cases with and without follower aerodynamics increase with the root angle of attack and the flow speed. However, follower effects have an overall much smaller impact on the flutter boundary than nonlinear kinematics, which is responsible for the variation in the flutter onset and offset speeds with the root angle of attack (or, equivalently, the tip vertical displacement).

Table 11 compares the tip vertical displacements at the hump mode flutter onset and offset speeds. Flutter occurs at almost the same tip vertical displacement with and without follower aerodynamics (0.5–0.7% difference in the analyzed root angle of attack range). The tip vertical displacements at the flutter offset speeds differ by about 4% but follow a similar trend. The flutter frequencies with and without follower aerodynamics, shown in Table 12, are practically the same. Differences remain below 1% for most of the flutter boundary, with 1.5–2% larger values only for the flutter frequencies at offset speeds for the highest roots angle of attack.

To complete the discussion, Fig. 16 shows root loci for $U = 0 \rightarrow 100$ m/s

Table 10: Hump mode flutter speed (m/s) with and without follower aerodynamics.

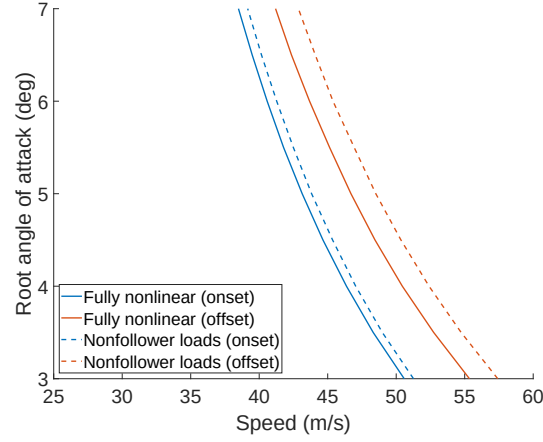
α_r (deg)	Onset speed			Offset speed		
	Follower	Nonfollower	Δ (%)	Follower	Nonfollower	Δ (%)
3.0	50.58	51.26	1.34	55.35	57.41	3.73
3.5	48.32	48.99	1.38	52.74	54.74	3.80
4.0	46.37	47.04	1.45	50.45	52.40	3.86
4.5	44.65	45.36	1.58	48.45	50.34	3.91
5.0	43.14	43.83	1.60	46.70	48.51	3.89
5.5	41.78	42.49	1.71	45.14	46.93	3.98
6.0	40.58	41.28	1.71	43.68	45.39	3.92
6.5	39.49	40.17	1.72	42.36	44.09	4.10
7.0	38.49	39.16	1.75	41.19	42.86	4.05

at $\alpha_r = 3, 5, 7$ deg. The aeroelastic eigenvalues with and without follower aerodynamics evolve along the same curves but achieve different values for each combination of flow speed and root angle of attack due to the different deformed shapes. These results highlight that the stability scenario of a very flexible wing at a given root angle of attack is driven by its deformation: the aeroelastic frequencies and damping values collapse onto the same curves when presented with the flow speed as an implicit parameter (in a root locus visualization or in frequency and damping diagrams with respect to the tip vertical displacement).

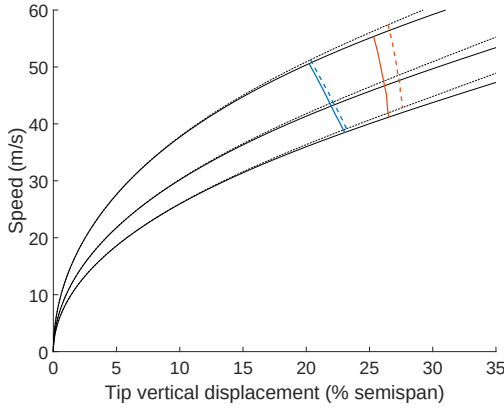
In summary, the hump mode flutter boundary is driven by nonlinear kinematics through its impact on the modal scenario, specifically, the reduced frequency separation between the structural modes that are originally the T1 and OOP2 modes in undeformed shape. Neglecting follower aerodynamics effects overestimates the flutter onset and offset speeds by up to 4% and predicts a wider hump mode instability region. However, these effects do not practically impact the flutter frequency nor change any qualitative trends for this wing.

4.5. Hump Mode Limit-Cycle Oscillations in Deformed Shape

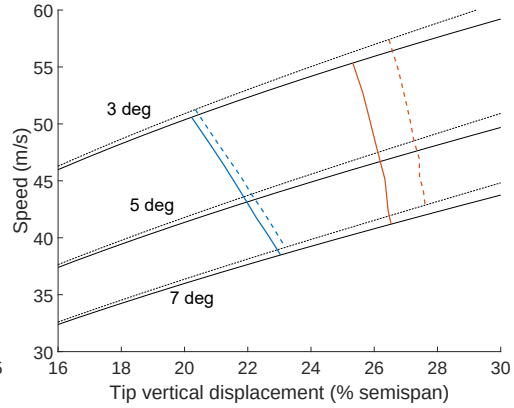
This section concludes the study by analyzing the impact of follower aerodynamics on the hump mode post-flutter dynamics at $\alpha_r = 5, 7$ deg, focusing on the amplitude of limit-cycle oscillations at the wing tip. Bifurcation dia-



(a) Varying root angle of attack



(b) Varying tip vertical displacement



(c) Zoom

Figure 15: Hump mode flutter boundary with and without follower aerodynamics.

Table 11: Hump mode flutter tip vertical displacements (% semispan) with and without follower aerodynamics.

α_r (deg)	Displacement at onset speed			Displacement at offset speed		
	Follower	Nonfollower	Δ (%)	Follower	Nonfollower	Δ (%)
3.0	20.22	20.34	0.57	25.32	26.46	4.50
3.5	20.78	20.88	0.49	25.64	26.76	4.38
4.0	21.25	21.36	0.53	25.85	26.97	4.32
4.5	21.63	21.79	0.70	26.03	27.13	4.25
5.0	21.97	22.12	0.65	26.19	27.26	4.07
5.5	22.27	22.44	0.80	26.34	27.43	4.13
6.0	22.56	22.72	0.73	26.39	27.43	3.94
6.5	22.81	22.96	0.66	26.44	27.56	4.22
7.0	23.04	23.19	0.66	26.53	27.61	4.05

Table 12: Hump mode flutter frequency (Hz) with and without follower aerodynamics.

α_r (deg)	Frequency at onset speed			Frequency at offset speed		
	Follower	Nonfollower	Δ (%)	Follower	Nonfollower	Δ (%)
3.0	30.51	30.74	0.77	28.12	28.08	−0.15
3.5	30.36	30.61	0.82	28.07	28.03	−0.12
4.0	30.23	30.48	0.85	28.04	28.02	−0.10
4.5	30.12	30.37	0.84	28.02	28.00	−0.08
5.0	30.01	30.28	0.90	28.00	27.99	−0.01
5.5	29.93	30.19	0.86	27.98	27.96	−0.06
6.0	29.84	30.10	0.88	27.34	27.99	2.38
6.5	29.75	30.02	0.92	27.47	27.77	1.10
7.0	29.68	29.95	0.93	27.60	27.18	−1.54

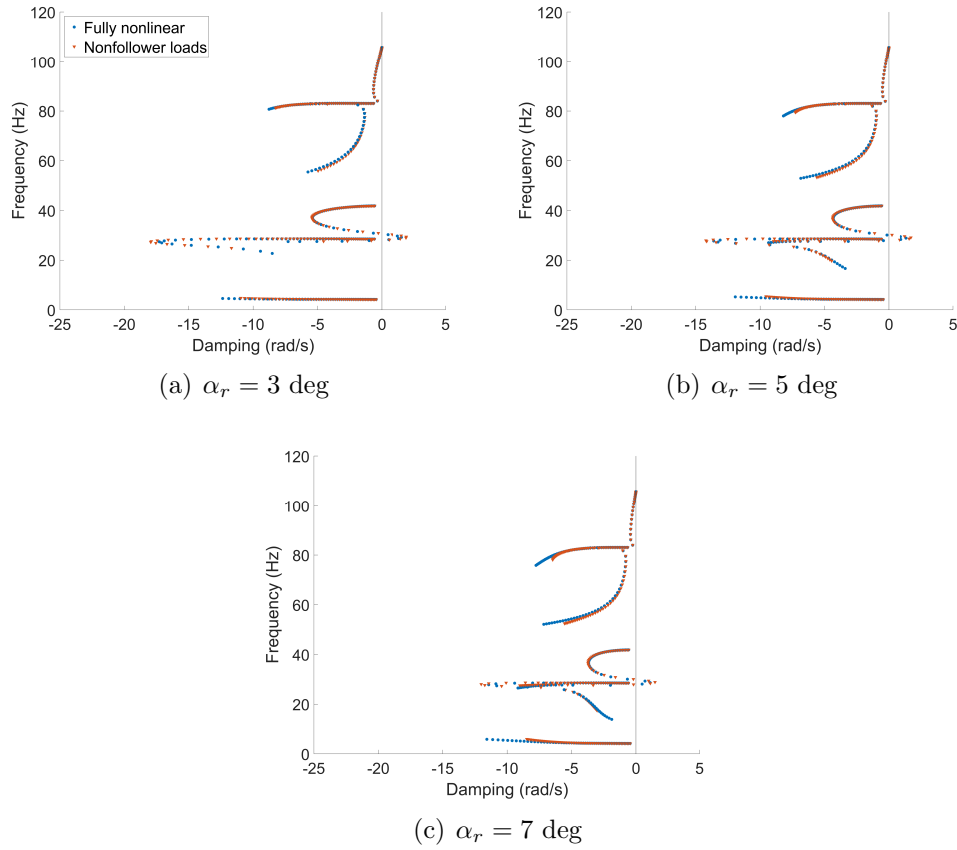


Figure 16: Root loci for $U = 0 \rightarrow 100$ m/s with and without follower aerodynamics.

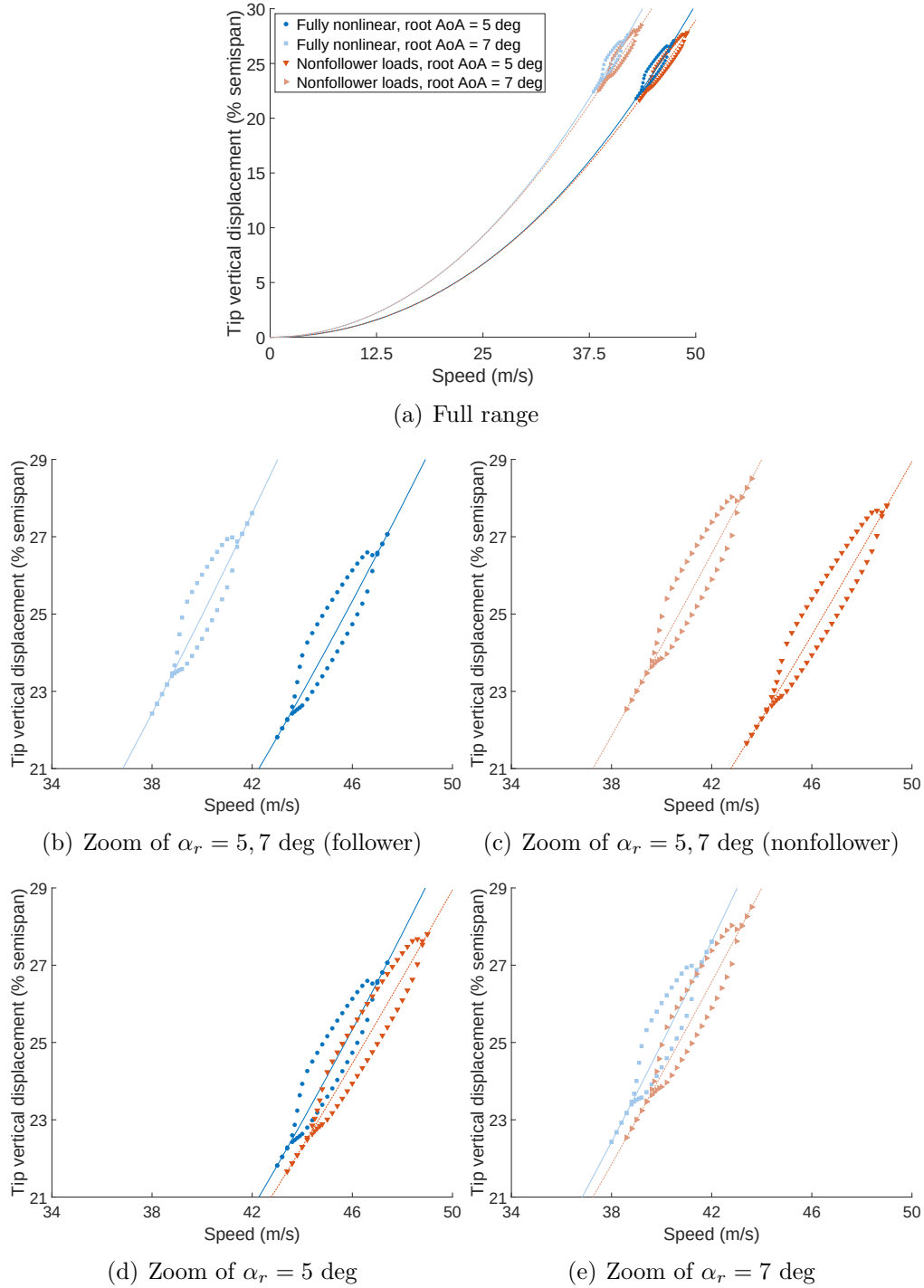


Figure 17: Hump mode bifurcation diagram for the tip vertical displacement with and without follower aerodynamics.

grams for these limit-cycle oscillations are obtained from direct time marching by extracting the maximum and minimum tip vertical displacements at the end of transient responses at different combinations of flow speed and root angle of attack. Each transient response involves perturbing the static aeroelastic deformed shape at a given combination of flow speed and root angle of attack with a unit tip force applied for a short time interval at the beginning of the simulation. Next, the fully nonlinear equations of motions given in Sec. 2 are marched in time until the transient response either recovers to the initial static deformed shape or it settles on a limit-cycle oscillation. The time marching is applied to the equations of motion in state-space form, where the states are the finite-element strain DOF, the corresponding strain rates, and the aerodynamic states.

Figure 17(a) shows the obtained bifurcation diagrams for the tip vertical displacement along with the static equilibrium paths (thin solid and dashed lines). The results with and without follower aerodynamics are denoted by different colors and markers, with higher root angles of attack in lighter shades. The bifurcation diagrams with and without follower aerodynamics are supercritical, as highlighted by the zooms in Figs. 17(a) to 17(d). These results indicate that the experimentally observed subcritical behavior associated with the Pazy wing hump mode is not driven by geometrical nonlinearities alone but rather by other nonlinear effects such as aerodynamic stall (and, potentially, by the interaction between structural and aerodynamic sources of nonlinear behavior). The flutter onset and offset speeds estimated from the transient responses are within 0.5 m/s of the results in Sec. 4.4 obtained from eigenvalue analyses. These slight differences are attributed to numerical damping in the time marching and to the long transients required to fully develop a self-sustained oscillation or to recover to the initial equilibrium close to the instability. Nonfollower aerodynamics shifts the bifurcation diagram at higher flow speeds while making it wider, in line with the flutter boundary results in Sec. 4.4. The bifurcation diagrams with and without follower aerodynamics are qualitatively similar and lie in approximately the same tip vertical displacement range of 22–28% semispan. Most of this range is associated with the static aeroelastic deflection corresponding to each combination of flow speed and root angle of attack. Dynamic oscillations in the tip vertical displacement about the geometrically nonlinear static equilibrium value are limited to small amplitudes of about 1–2% semispan. The small-amplitude dynamic oscillations observed in this numerical study are consistent with observations from Pazy wing flutter tests

(although the bifurcation diagram for the tip vertical displacement or any other output metric was not fully characterized experimentally). Drachinsky et al. [13] observed that “the flutter onsets showed large wingtip twist angles of over 10 deg in the flutter mode compared to deformation of only a few percent of the span, suggesting that the primary source of nonlinearity in the oscillating system is aerodynamic stall”. The slight impact of follower aerodynamics on the nature and amplitude of limit-cycle oscillations in Figure 17 is attributed to the small dynamic excursions about each static aeroelastic deformed shape, which translate to small changes in the magnitude and orientation of the aerodynamic loads with respect to their distribution at the geometrically nonlinear equilibrium.

The bifurcation diagrams show that the limit-cycle oscillations are not symmetric about the statically deformed shape. The static tip vertical displacements remain within the amplitude range of the bifurcation diagram, but their values are below the averages of oscillation maxima and minima except for conditions very close to the flutter onset speeds. This asymmetric behavior was observed in previous studies on very flexible wings [4].

In summary, follower aerodynamic loads shift and shrink the hump mode bifurcation diagram due to the lower flutter onset and offset speeds, but do not change qualitative trends for this wing. Considering only geometrical nonlinearities predicts a supercritical behavior in place of the experimentally observed subcritical one [13], which is attributed to stall effects and their interaction with large deflections.

To conclude the discussion, Fig. 18 shows time histories for the tip vertical displacement at the root angle of attack $\alpha_r = 5$ deg and flow speed $U = 46$ m/s for different values of the tip force used to excite the wing. These time histories are included to show that the obtained limit-cycle oscillations are independent of the initial perturbation. For conciseness, the results are only shown for the case of follower aerodynamics as the results for nonfollower aerodynamics have the same trends. While different tip forces $F_z = 0.5, 1, 5, 10$ N result in different transients, the dynamic response settles on the same limit-cycle oscillation. The only differences among the results for increasing values of the applied tip force is that the initial transient involves larger amplitudes and lasts a shorter time, as limit-cycle oscillation develop faster for larger perturbations. The different time duration of the initial transient results in a phase shift between the limit-cycle oscillations for different values of the applied tip force with no change in their amplitude and frequency, as evidenced in Fig. 18(b).

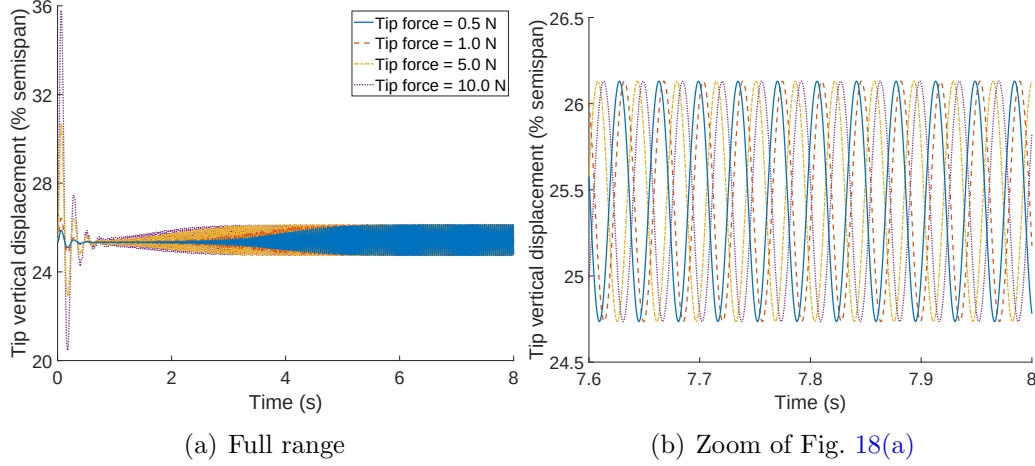


Figure 18: Time histories for the tip vertical displacement at $\alpha_r = 5$ deg and $U = 46$ m/s with follower aerodynamics for different initial perturbations.

5. Concluding Remarks

This paper investigated geometrically nonlinear effects in wing aeroelastic dynamics at large deflections in low-speed flow, using the Pazy wing as the test case representative of a very flexible wing. The study leveraged a verified and validated low-order model of the Pazy wing composed of a geometrically nonlinear beam coupled with potential flow thin airfoil aerodynamics. The complete model was implemented into the University of Michigan’s Nonlinear Aeroelastic Simulation Toolbox (UM/NAST), a low-order multidisciplinary framework for studying very flexible wings and aircraft. The framework can consider geometrical nonlinearities associated with nonlinear kinematics and follower aerodynamic loads individually or in combination.

The results showed that nonlinear kinematics is the primary geometrical nonlinearity influencing the wing static aeroelastic response, natural frequencies in deformed shape, and hump mode flutter mechanism. Nonfollower aerodynamics underestimated tip vertical displacements by up to 10%, consequently predicting 1–4% higher hump mode flutter onset and offset speeds due to the lower decrease in the frequency of the torsion-dominated mode with the flow speed. However, the modal and stability scenarios with and without follower aerodynamics did not qualitatively change and the results for a given root angle of attack were found to evolve along the same curves once plotted as functions of the tip vertical displacement, with the flow speed

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as an implicit parameter. Similarly, follower aerodynamics shifted the hump mode bifurcation diagram for the tip vertical displacement to lower flow speeds and shrank its flow speed bounds with no change the bifurcation behavior or amplitude of limit-cycle oscillations. Predictions of the hump mode instability at low deflections were shown to strongly depend on the modeling of three-dimensional *unsteady* aerodynamic effects. At larger deflections, the modeling of three-dimensional *steady* aerodynamic effects played a key role due to its impact on the predicted static deformed shape and resulting structural frequencies. Geometrically nonlinear effects alone missed the subcritical nature of the hump mode flutter mechanism, which was rather attributed to aerodynamic stall effects and their interaction with large deflections.

This study contributes to the growing literature on aeroelastic dynamics of very flexible wings by investigating geometrically nonlinear effects for a wider deflection range than considered in previous work and by assessing their impact on a hump-mode-type flutter mechanism, including the resulting limit-cycle oscillations. To ease future comparisons, all the results from this work is publicly available for download from the authors’ GitHub repository², which already contains the low-order model properties and the results from previous verification and validation studies. Together with other efforts by the Third Aeroelastic Prediction Workshop Large Deflection Working Group, this study aims to advance our fundamental understanding and predictive capabilities for very flexible wings to support the design of future energy-efficient aircraft.

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